Speech and Audio Processing

05 - speech recognition

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Class schedule (tentative)

01	Oct. 02	Introduction
02	Oct. 09	Speech production / 音声生成
03	Oct. 16	Speech analysis and deep learning / 音声分析と深層学習
04	Oct. 23	Speech synthesis / 音声合成
05	**Oct. 30**	**Speech recognition / 音声認識**
06	Nov. 6	**Guest Talk (acoustic signal processing) / ゲストトーク (音響信号処理)**
07	Nov. 13	Foundation model / 基盤モデル
08	Nov. 27	Advanced topics / 発展的話題
09	Dec. 04	**Guest Talk (music information processing) / ゲストトーク (音楽情報処理)**
10	Dec. 11	Speech system building / 音システム構築
11	Dec. 18	Speech system building / 音システム構築
12	Dec. 25	Speech system building / 音システム構築
13	Jan. 08	Speech system building / 音システム構築
14	Jan. 22	Final presentation / 最終発表

Last week's class: text-to-speech (TTS)

What happens in computers when artificially **synthesizing** speech



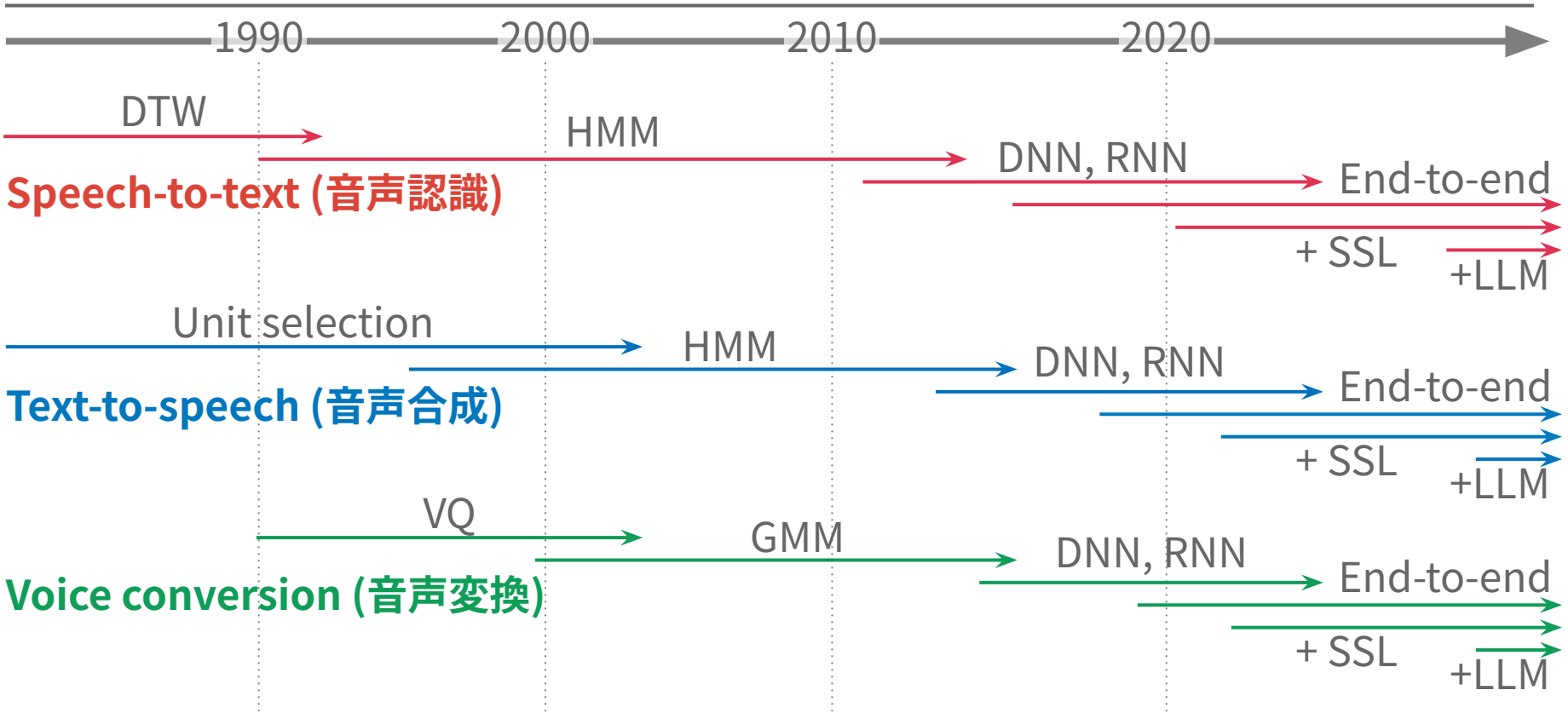
Today's class: automatic speech recognition (ASR)

What happens in computers when artificially **recognizing** speech



Review of last week’s content / 前回の復習

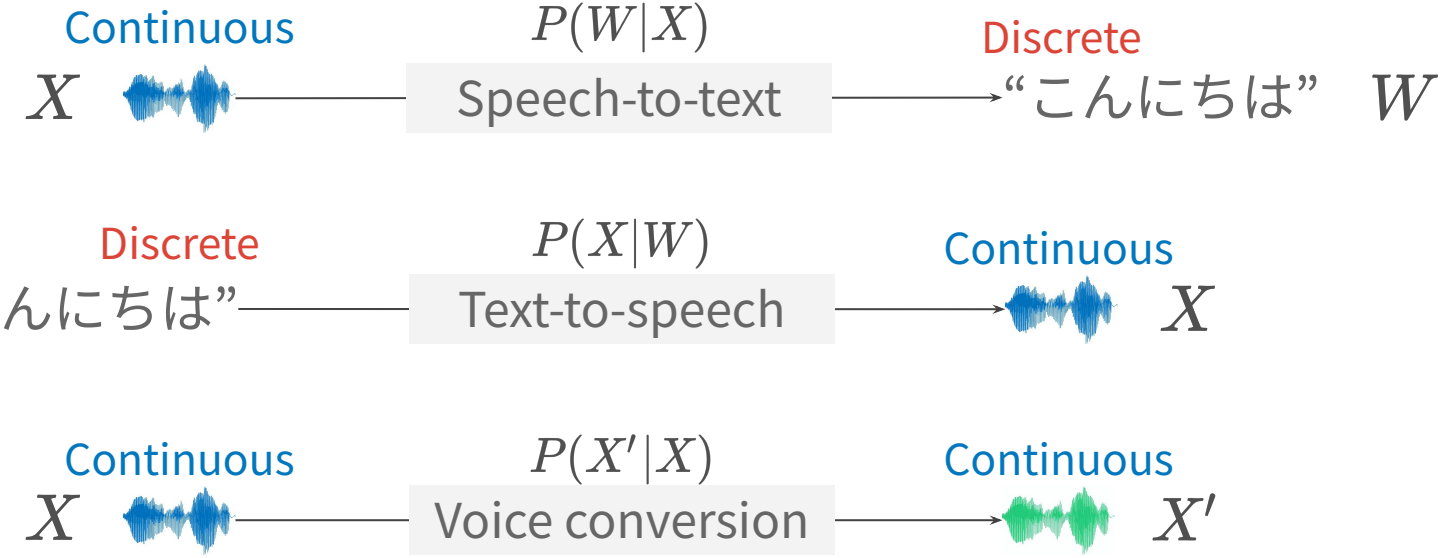
Data driven approach to TTS



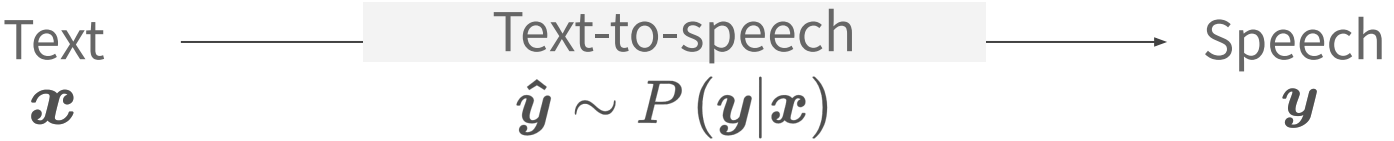
* DTW: dynamic time warping, HMM: hidden Markov model, DNN: deep neural network, RNN: recurrent neural network
VQ: vector quantization, GMM: Gaussian mixture model, SSL: self-supervised learning, LLM: large language model

Formulation / 定式化

Variable-length conversion from a {continuous, discrete} sequence to another {continuous, discrete} sequence.



Formulation / 定式化

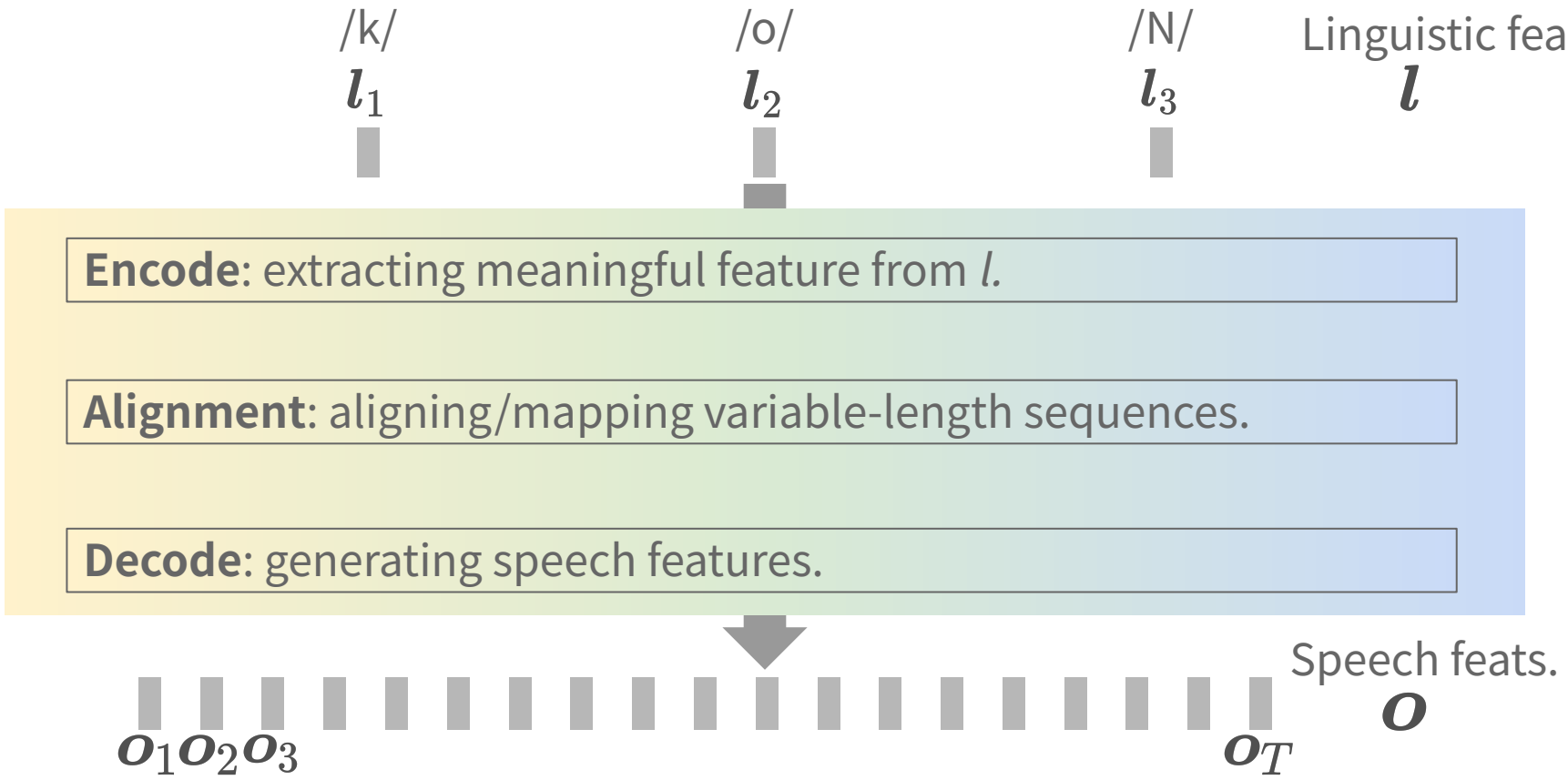


Why are statistical models used?

- Speech production is probabilistic
  - Motor command: It changes even if you pronounce the same word.
  - Radiation: Frequency phase varies when you move.

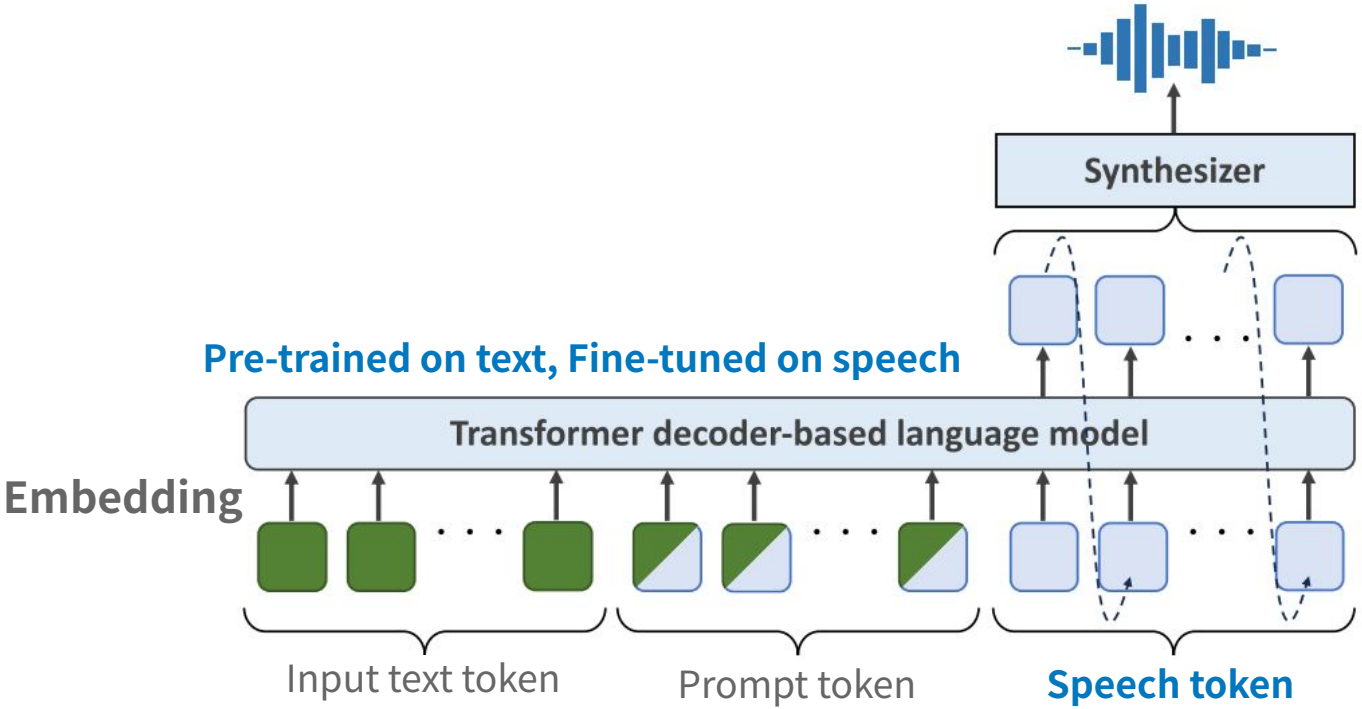
- Pronunciation of text is not deterministic
  - Homophone (同音異義語)：移動 (i do u) vs. 異同 (i do u)
  - Heterophone (同形異音語)： 市場 (shi jo u) vs. 市場 (i chi ba)

Converts linguistic features to speech features



Typical LLM-based TTS

Predicts next speech token using a LLM.



https://www.jstage.jst.go.jp/article/jasj/81/10/81_719/_pdf/-char/ja

History / 歴史

Audrey from Bell Lab (1952): 10 digits

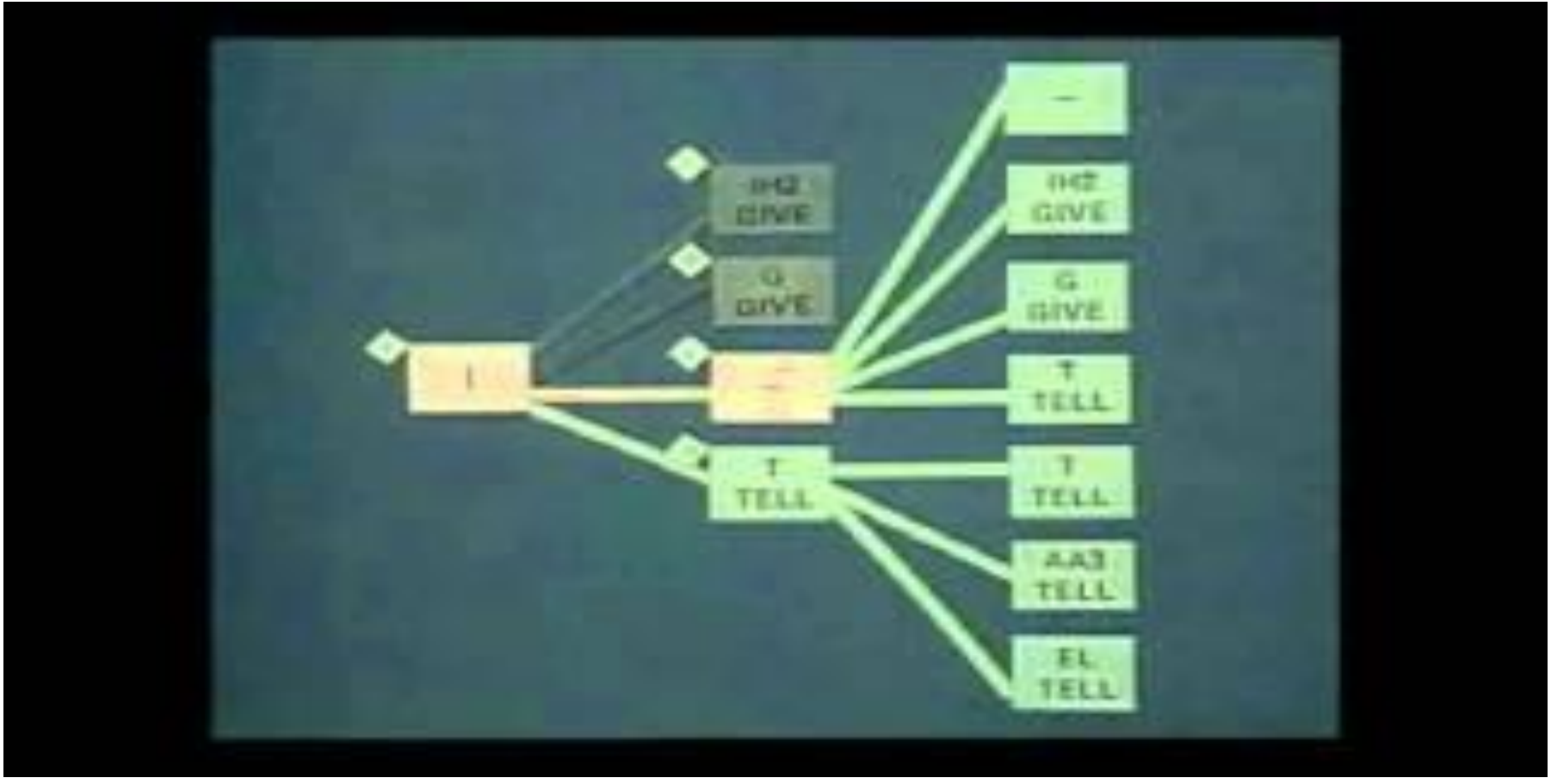


https://computerhistory.org/blog/audrey-alexa-hal-and-more/

Phonetic typewriter (1962): 100 Japanese syllables

L1. Phonetic Typewriter. T. SAKAI (nonmember) AND S. DOSHITA (nonmember), *Kyoto University, Kyoto, Japan.*—We have constructed a research model of phonetic typewriter named Sonotype since 1960. This equipment was designed based upon our research results of these several years, aiming at both universal and flexible ability in the processing of speech sound. The basic equipment was a device for one hundred Japanese monosyllables in which about 3000 transistors and about 5000 diodes were used. For the purpose of treatment of Japanese conversational speech, some parts of control system have been added considering the data obtained from the basic equipment; however, the experimental result has not yet been obtained. This equipment has several peculiarities together with some features similar to those reported in other papers. (a) All the control pulses are derived from the input speech sound wave itself. (b) Distinctive features of speech sound are obtained by using not only the results of ordinary spectrum analysis but also various kinds of features necessary for the recognition of phonemes: for instance, voiced, unvoiced, fricative, nasal, contracted, and so on. (c) Discrimination is made by digital technique such as order pulse, AND gate, OR gate and binary circuit. Of course it is necessary to convert analog quantity to digital one and vice versa.

CMU Harpy: ~ 1000 words



ViaVoice from IBM, based on machine learning



Deep learning era since 2010s



Nowadays



https://www.biccamera.com/bc/i/topics/osusume_smart_speaker/index.jsp?srsltid=AYCX9vK_pVyemfcdArhPgpkU9S_2CKJbG4rNDFLZVXLxY1CyylaenbsH



https://www.meta.com/jp/ai-glasses/ray-ban-meta/?srsltid=AYCX9vKsMblHdny_UHAB9R9I7yUu4q7aw2b1LT6flJ92MO4AM2WfbLJV

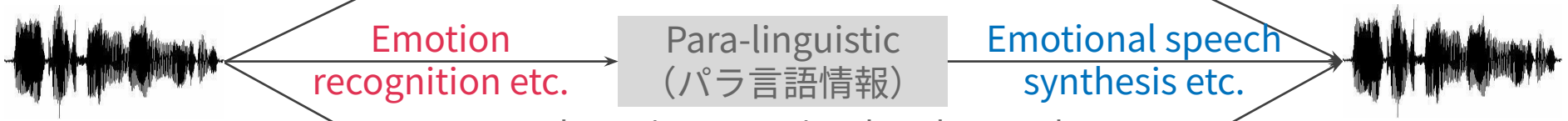


https://www.advanced-media.co.jp/en/newsrelease/21502/

Mathematical formulation / 定式化

Speech information

Physical → **Information** → **Physical**



Speech-to-text

Linguistic
（言語情報）

Phonetic properties

Text-to-speech

Emotion
recognition etc.

Para-linguistic
（パラ言語情報）

Non-phonetic properties that the speaker consciously expressed (e.g., emotion).

Emotional speech
synthesis etc.

Speaker
recognition etc.

Non-linguistic
（非言語情報）

Non-phonetic properties that the speaker unconsciously expressed. (e.g., individuality)

Multi-speaker
speech synthesis etc.

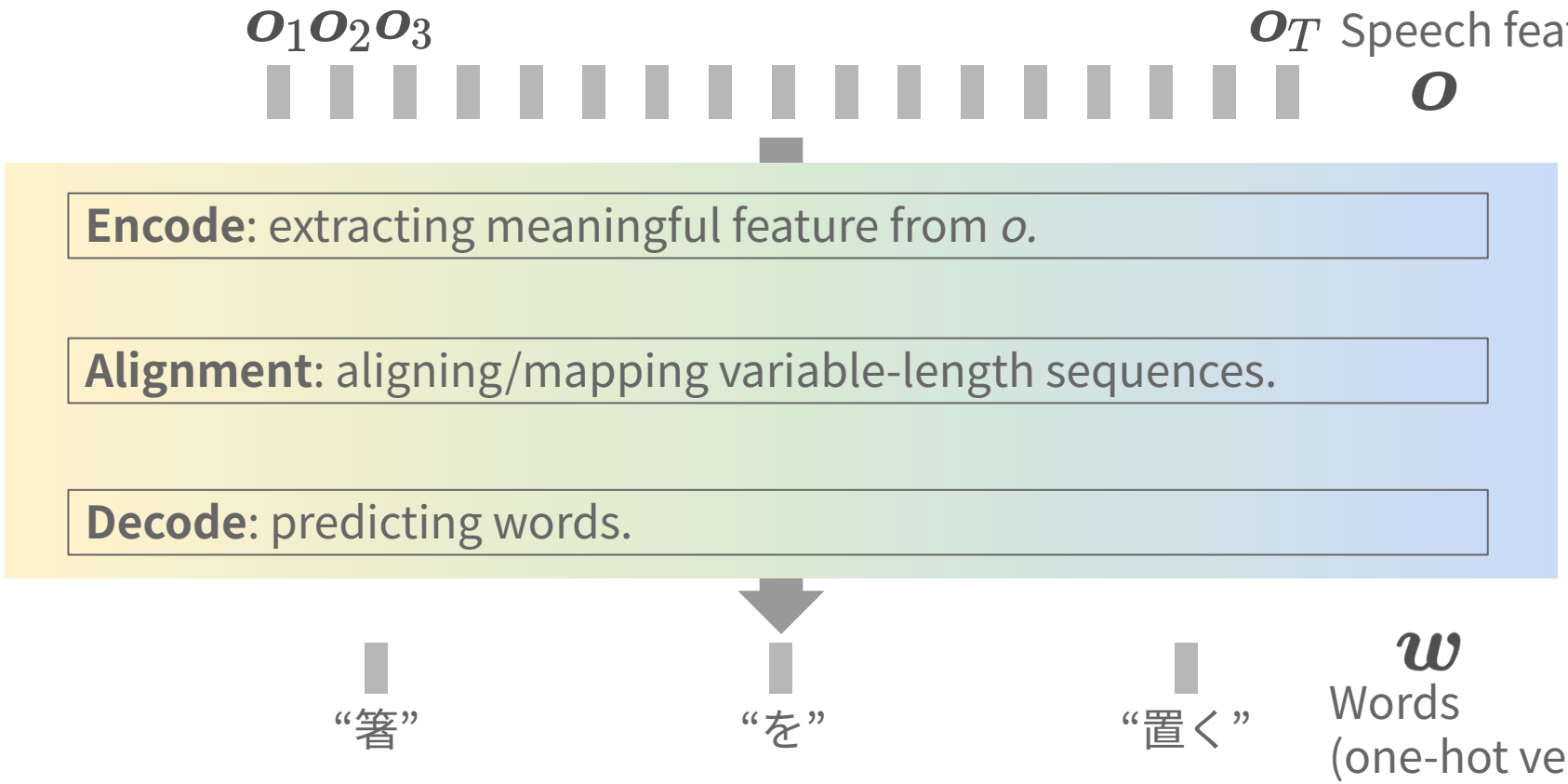
Formulation / 定式化



Must be robust to:

- Paralinguistics: emotion, mental state
- Non-linguistics: speaker, disease, accented language, etc.
  - From children to elderly
  - From healthy speech to pathological speech
  - From native speaker to language learner
    - straight (/s t r e i t/) vs. ストレート (/s **u** t **o** r e i t **o**/)
- Channel: backgrounds, telecommunication, reverberation, etc.
  - From living room to festival venue

Inner modules



Alignment & Decoder

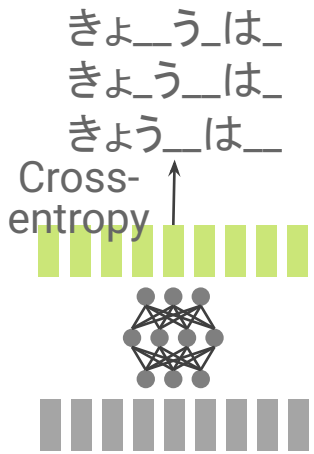
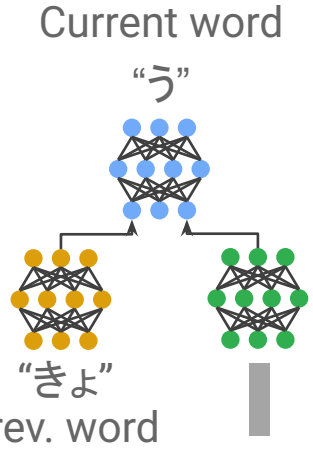
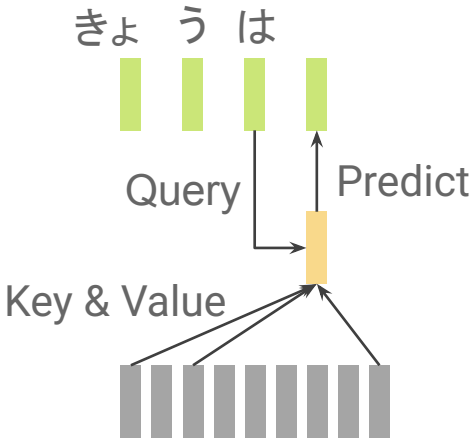
- **Alignment: Soft vs. hard**
  - Soft alignment: hard computation (O(N^2)) but rich feature
  - Hard alignment: poor feature but light computation (e.g., streaming)



- **Decoder: Autoregressive vs. non-autoregressive**
  - Autoregressive: sequential computation but good prediction
    - Current word can be predicted from past predicted words.
  - Non-autoregressive: poor prediction but parallel computation (i.e., fast)



Three types of end-to-end model

	CTC (connectionist temporal classification)	Transducer	LAS (listen, attend, and spell)
Fig.	 Use blank mark "_" to align sequences.	 Recognize a word using prev.-recognized words.	 a.k.a., source-target attention
Pros.	HMM-like lightweight streaming process	Dependency among words & streaming process	Dependency among words & well-known technique
Cons.	Independence among words	Heavy computation in training (much memory)	Sequence-wise process

CTC (connectionist temporal classification)

Introducing “<blank>” (or “_”, “<b>”) symbol

- blank = “no label”
  - Inserted so that the word (or phoneme) sequence has the same length to speech feature sequence
- Example (using phoneme but same when using words)
  - Assuming speech feature length is 8.
  - Extended sequence is designed to uniquely reconstruct the original.

<b>	i	i	i	<b>	a	<b>	i
i	<b>	<b>	a	a	<b>	<b>	i
i	i	<b>	a	<b>	i	i	<b>

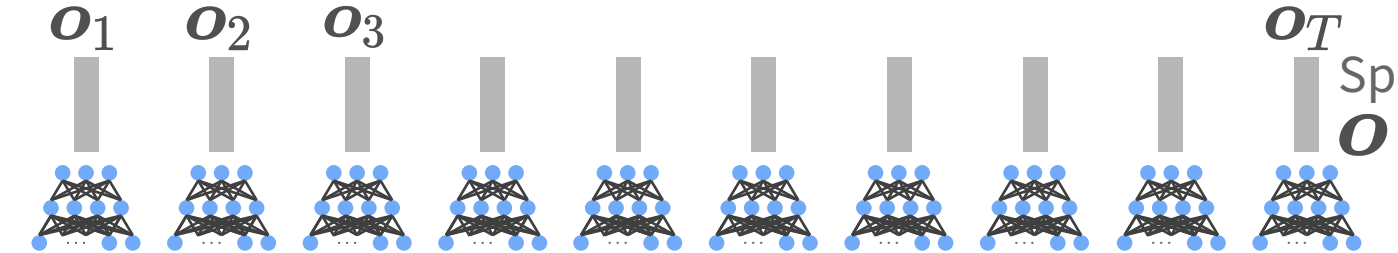
deduplicate consecutive non-blank

delete blank

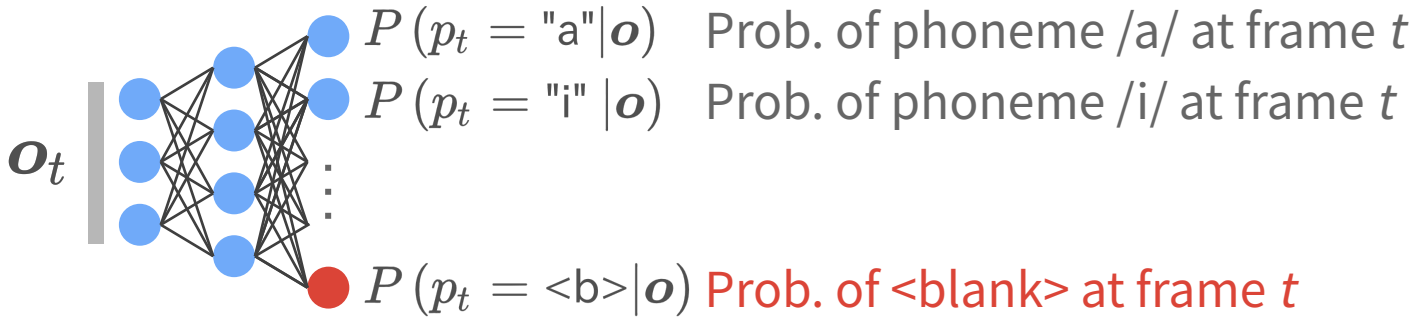
/i/ /a/ /i/ (original)

Output layer for CTC

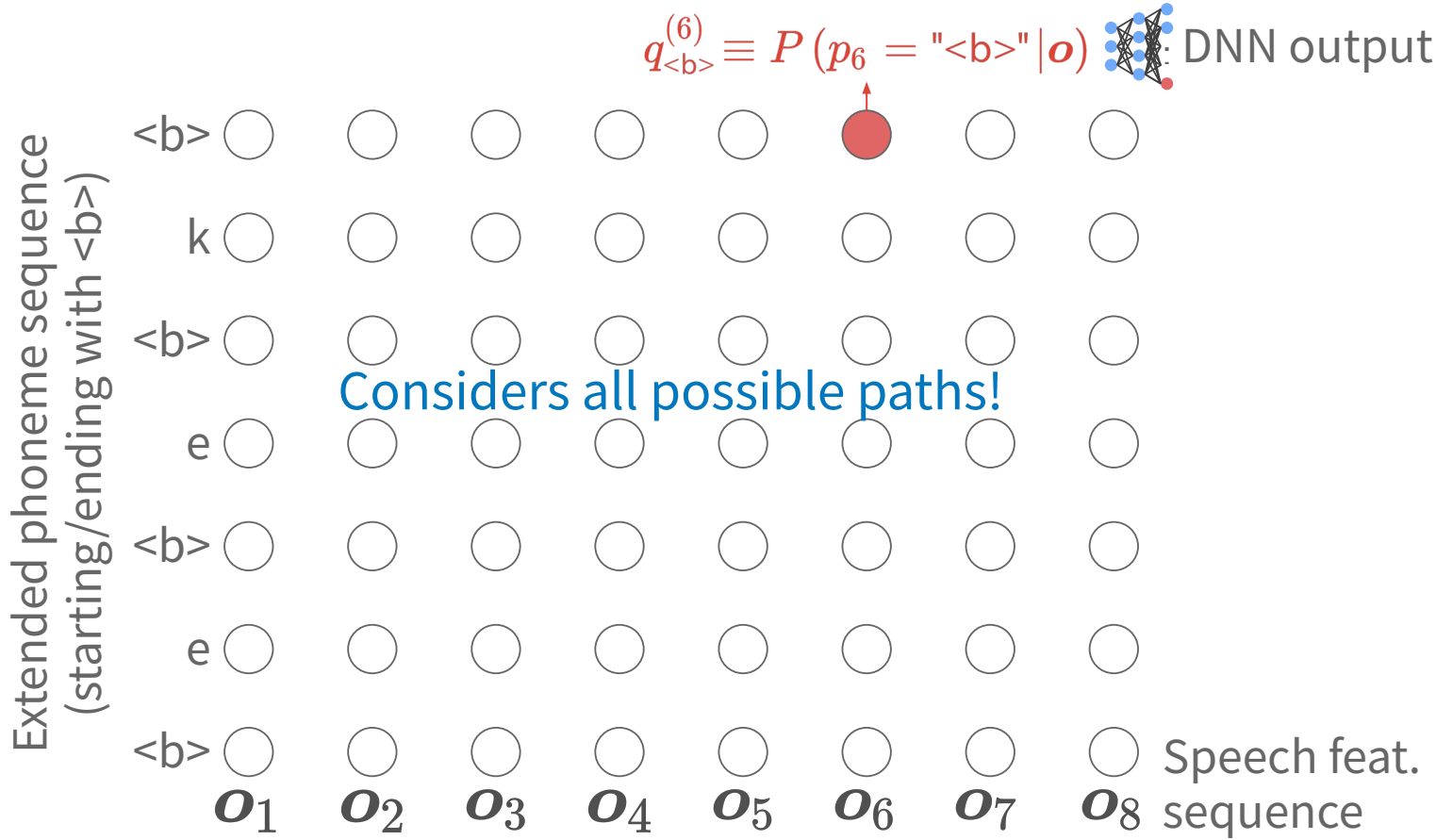
Architecture



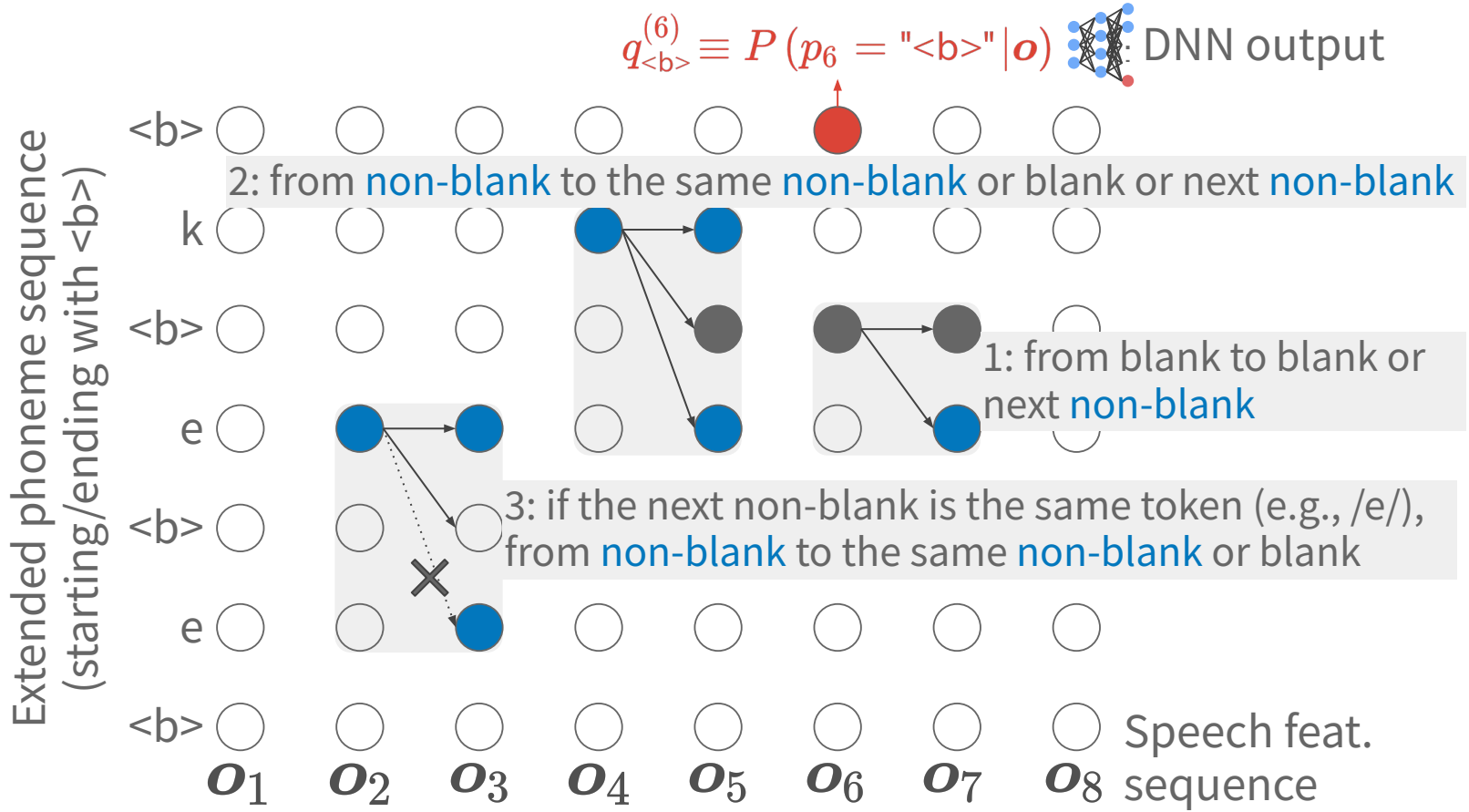
Output: phoneme set (or word set) + {blank}



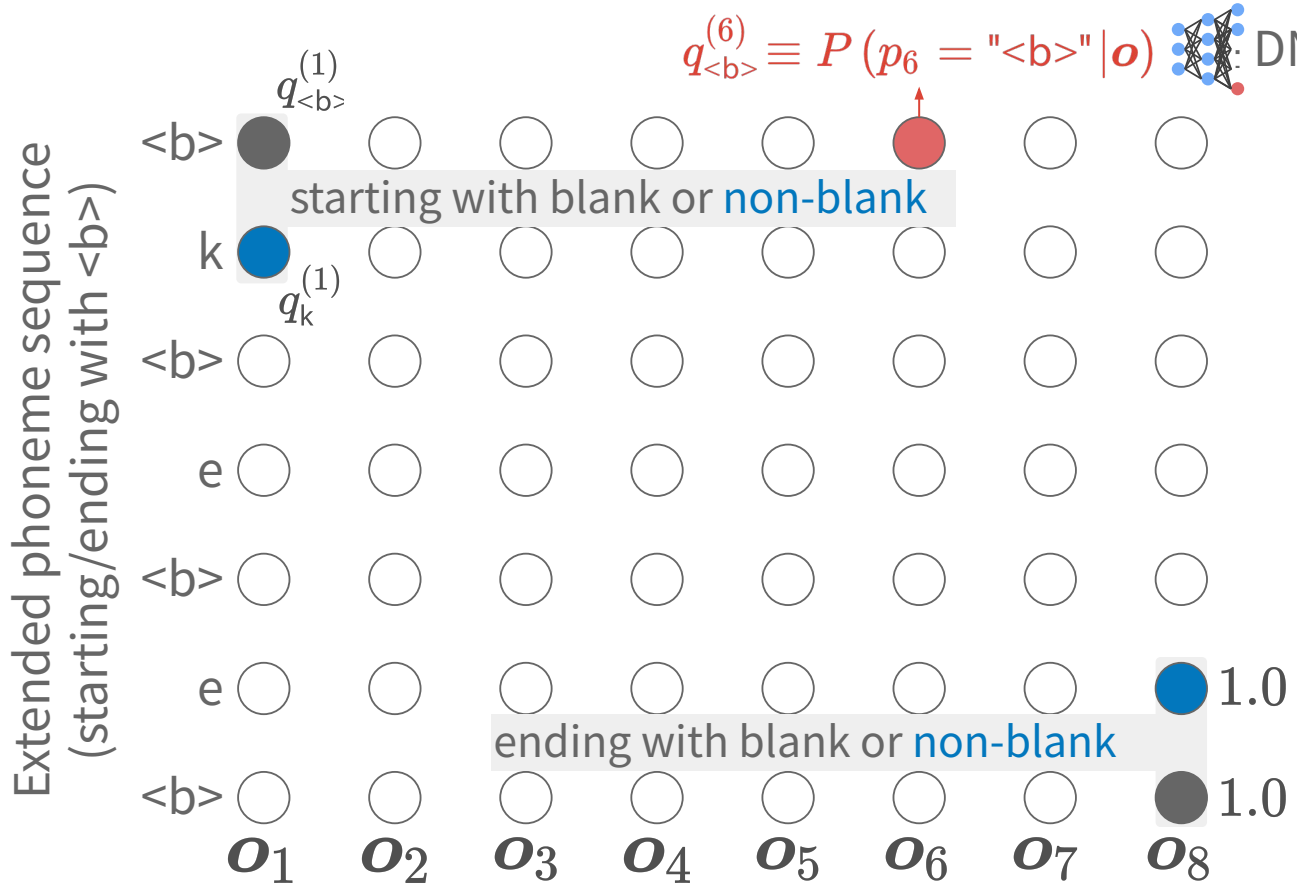
Alignment between speech feature sequence and extended phoneme sequence



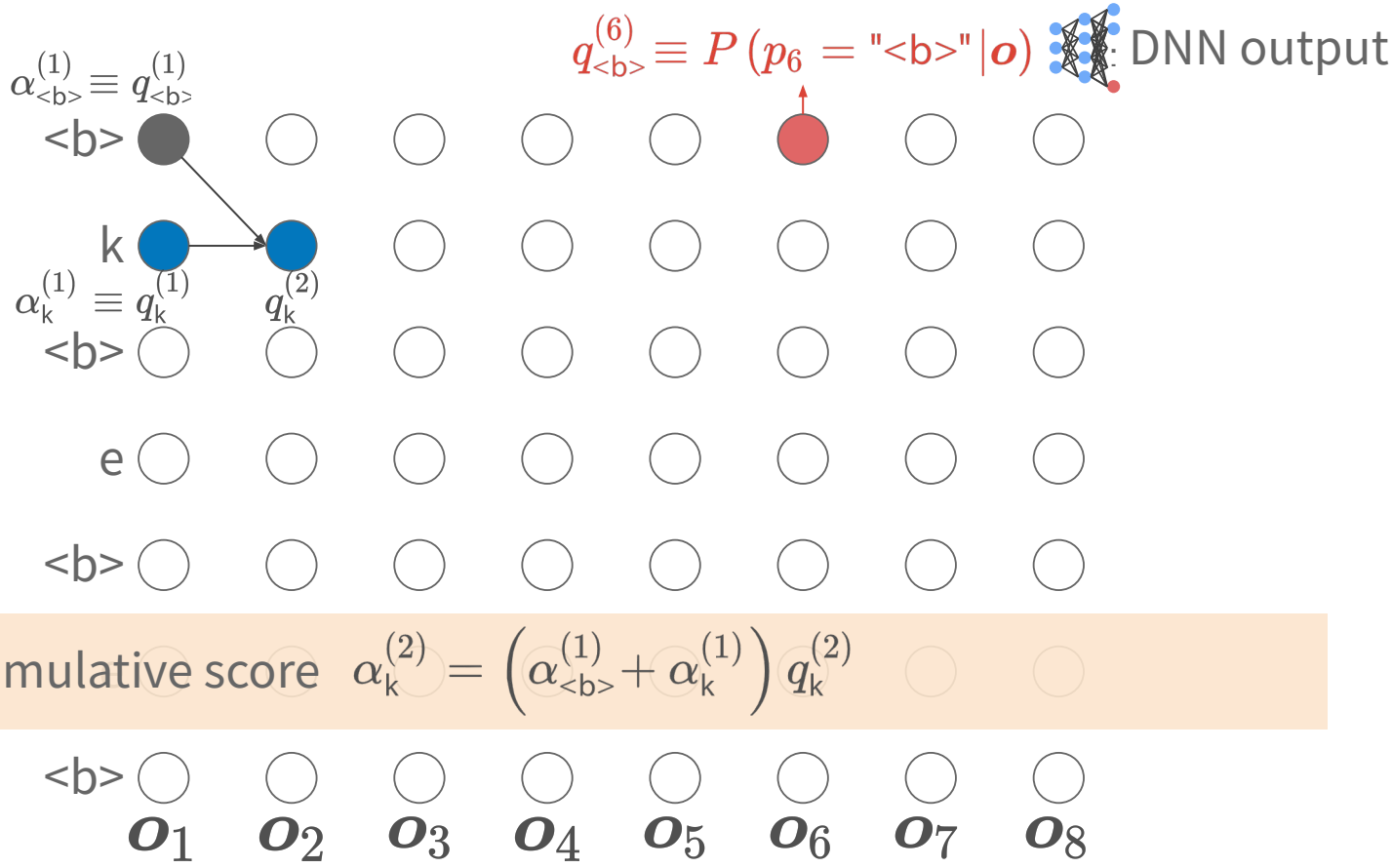
Transition conditions



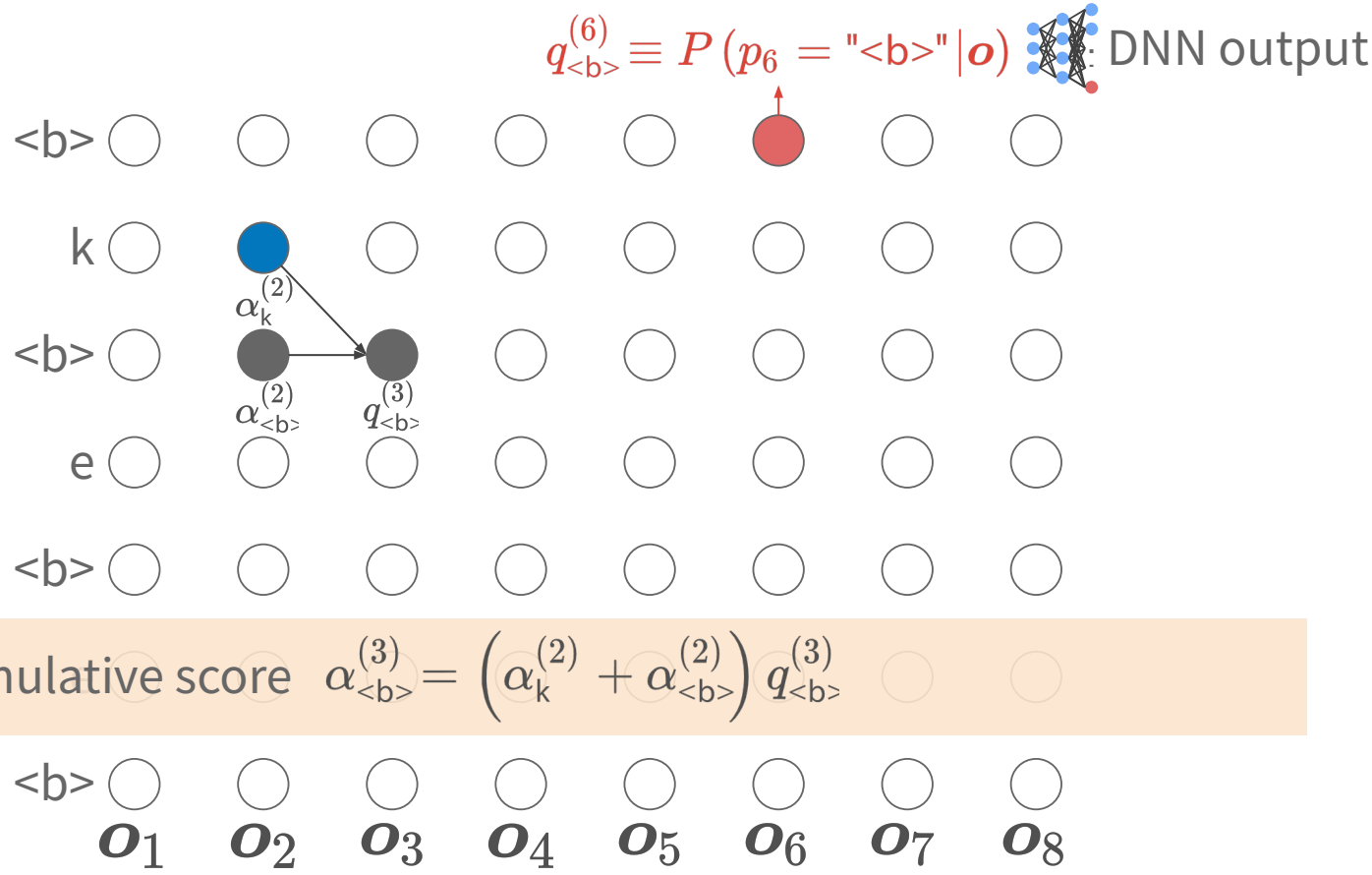
Start-end conditions



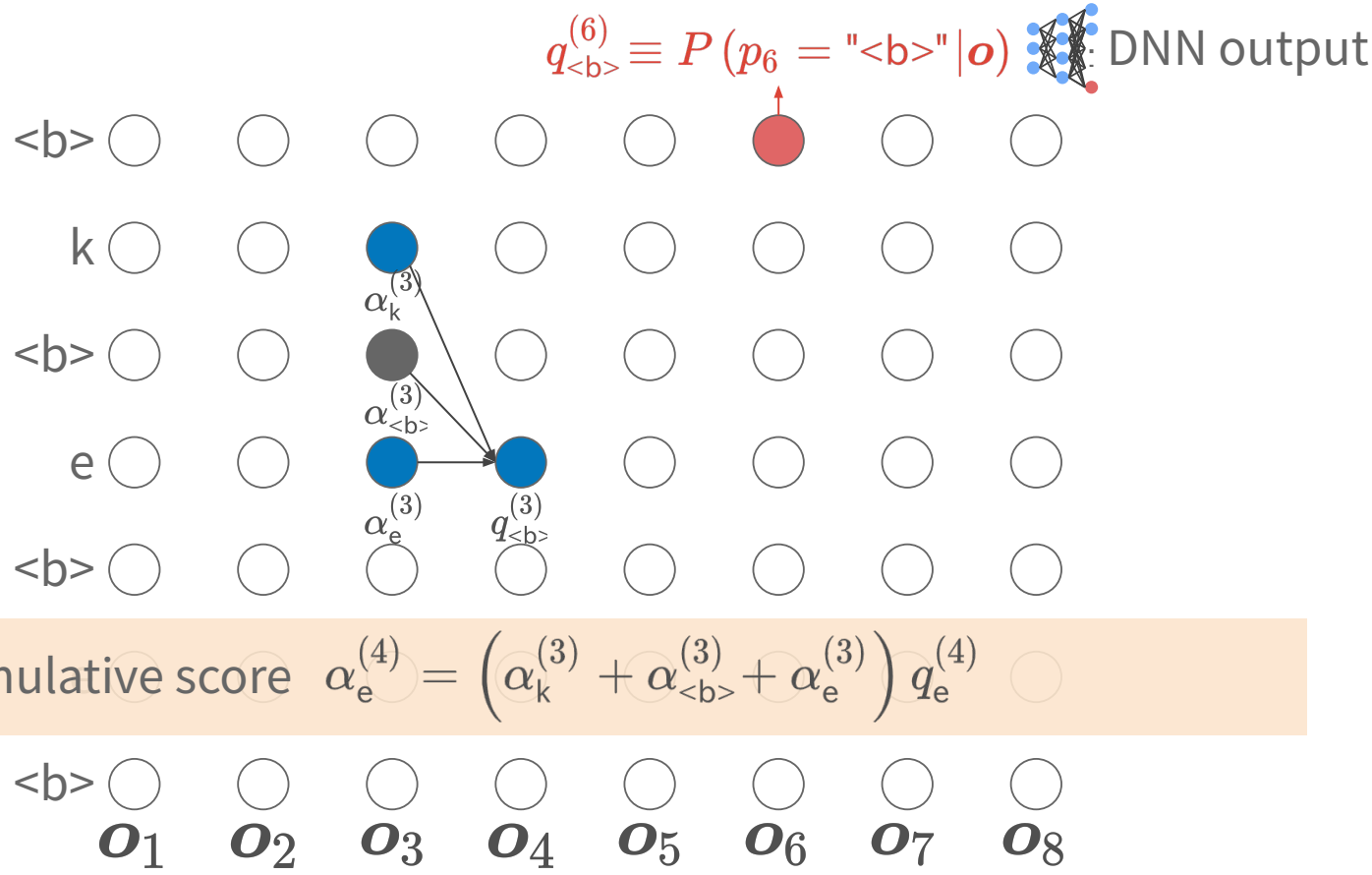
Forward transition



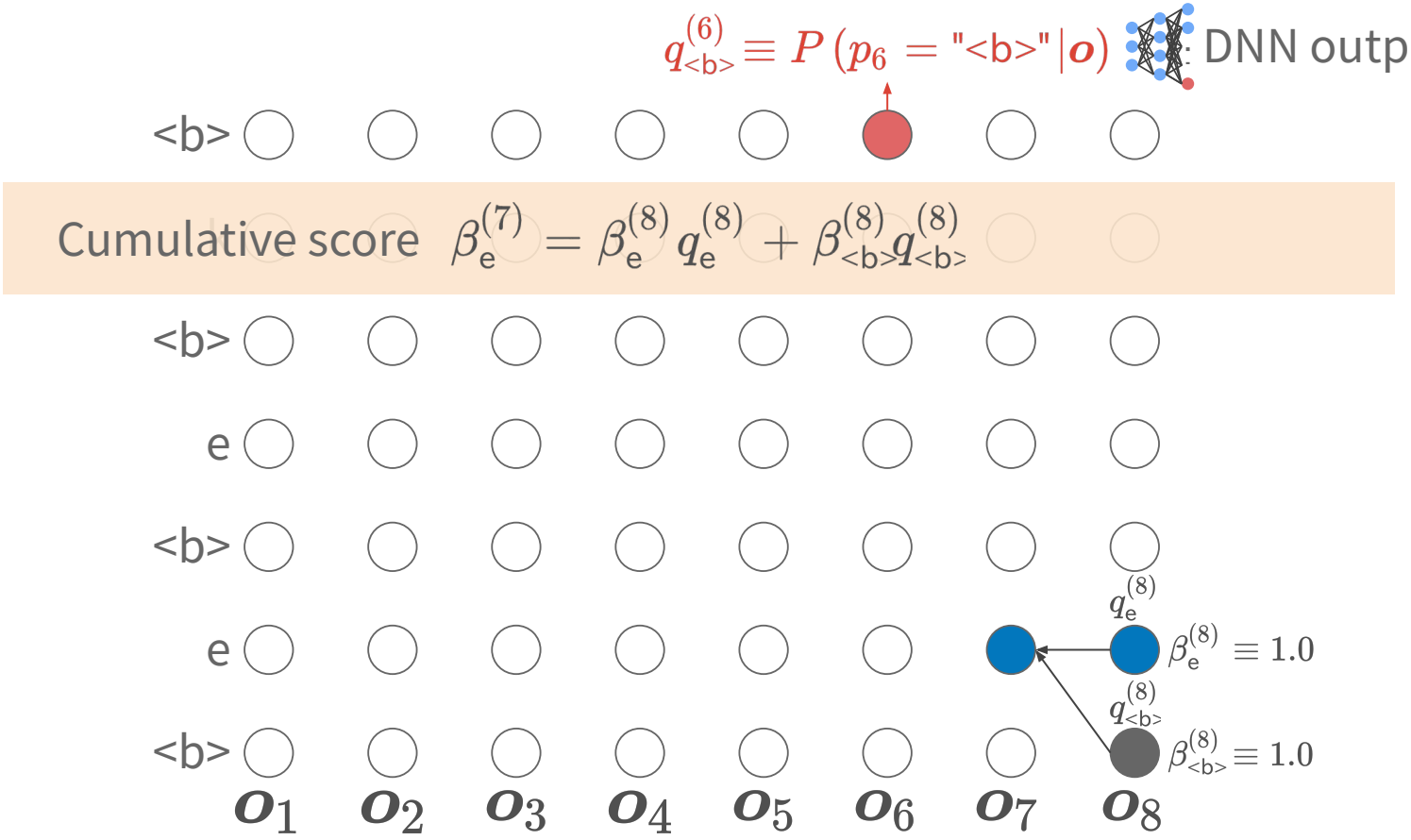
Forward transition



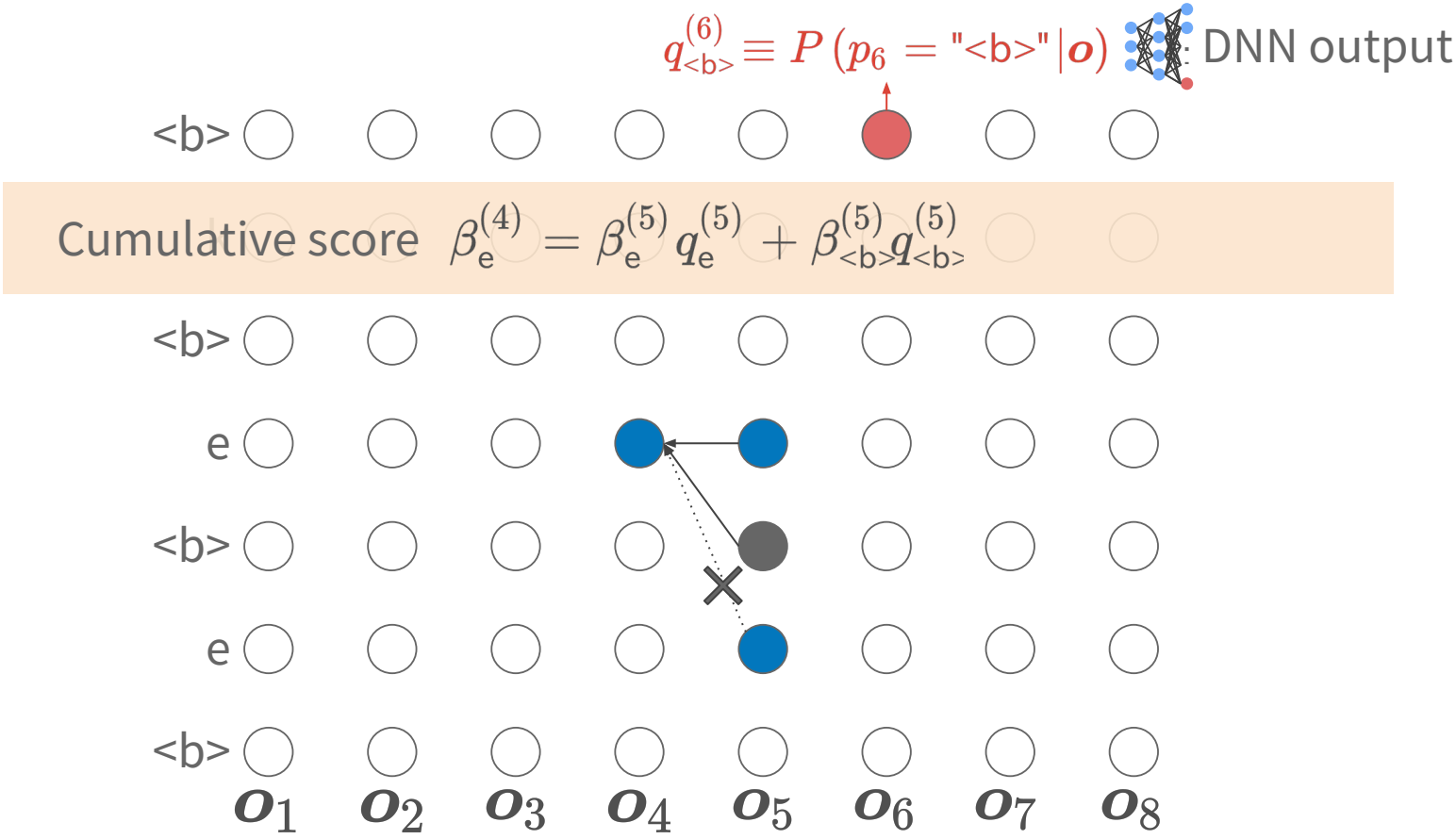
Forward transition



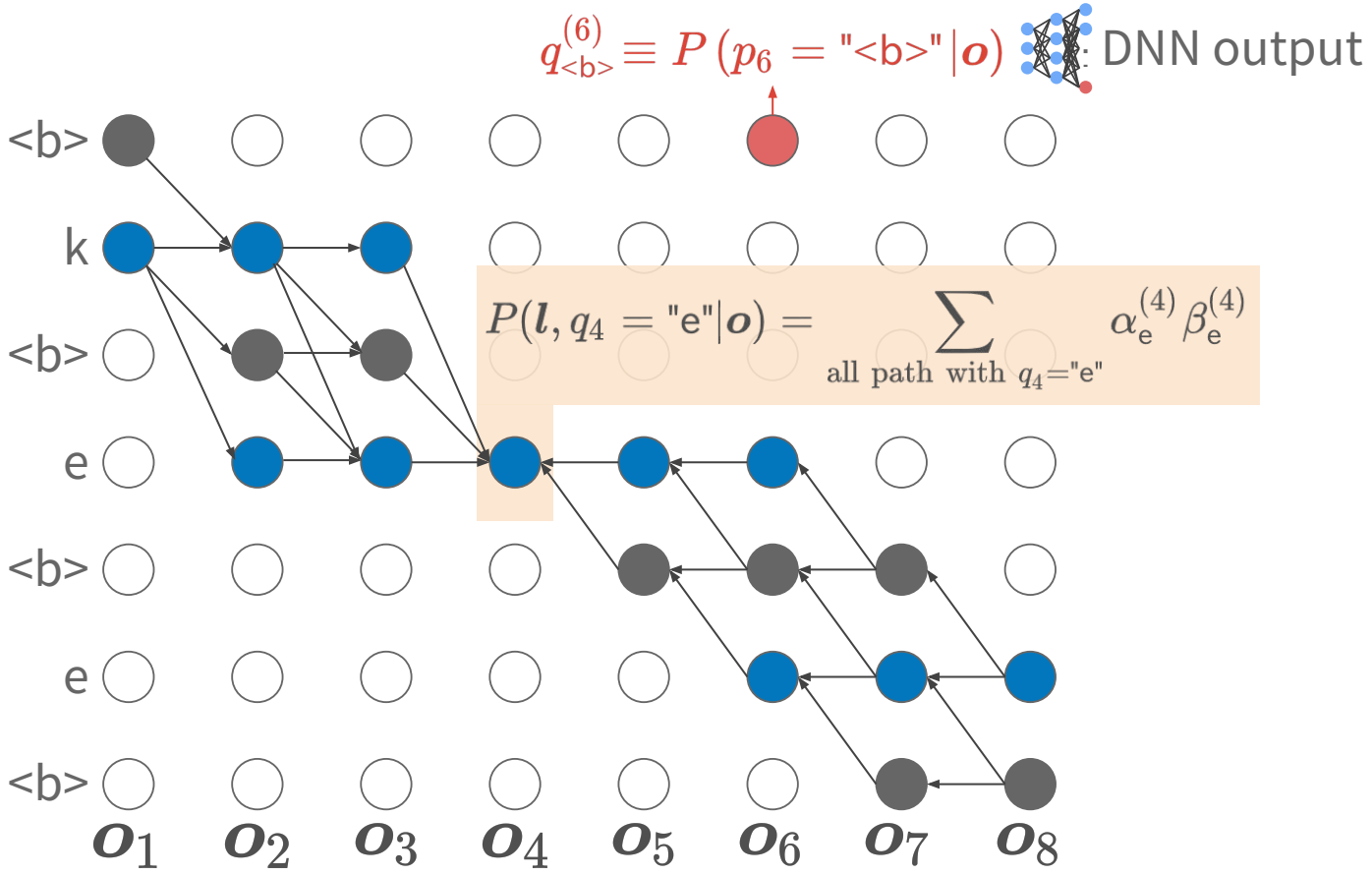
Backward transition



Backward transition



Possibility of a phoneme at a frame is calculated by marginalizing all possible paths with the phoneme.



Training and inference

- Training: Considering all possible paths.

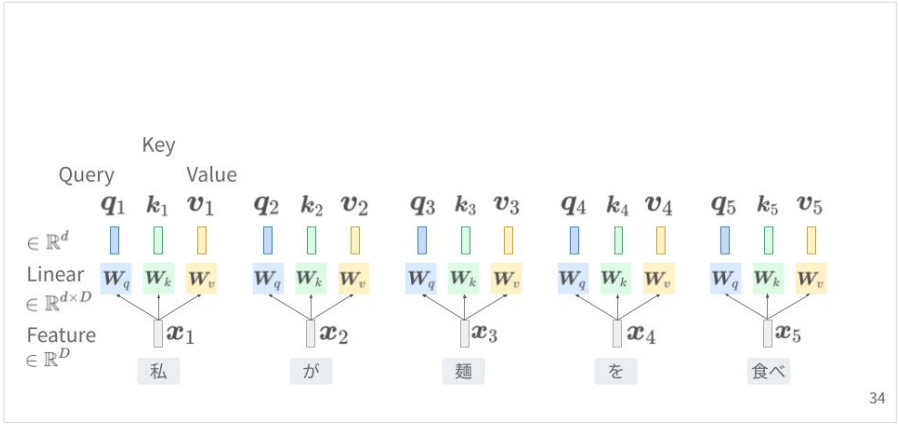
$$\mathcal{L}_{\mathrm{CTC}} = -\log P\left(\boldsymbol{l}|\boldsymbol{o}\right)$$

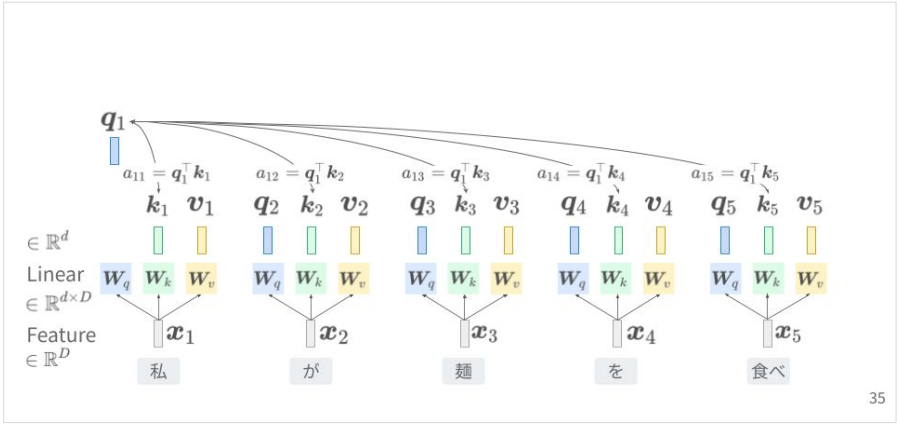
- Inference: Pathfinding
  - Greedy search:
    - Find the most possible phoneme or blank at each frame.
    - Convert the extended phoneme sequence to no-blank sequence.
  - Beam search:
    - One kind of breadth-first search (幅優先探索)
    - Keep N-best possible paths and determine the best.

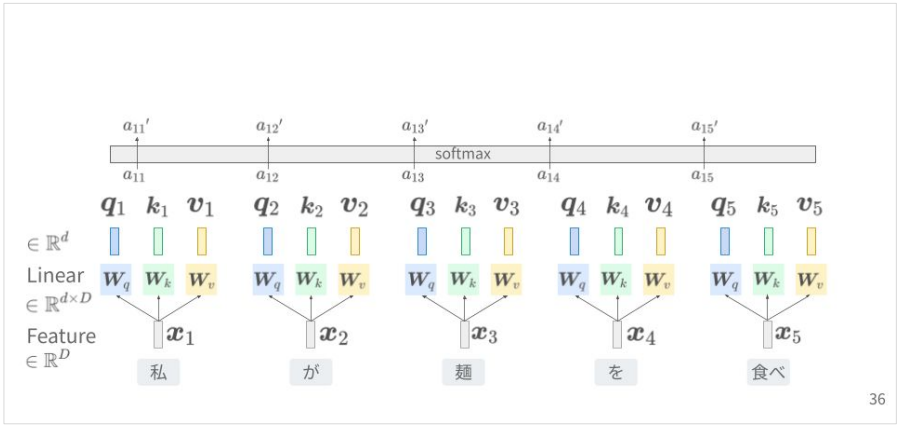
LAS (listen, attend and spell)

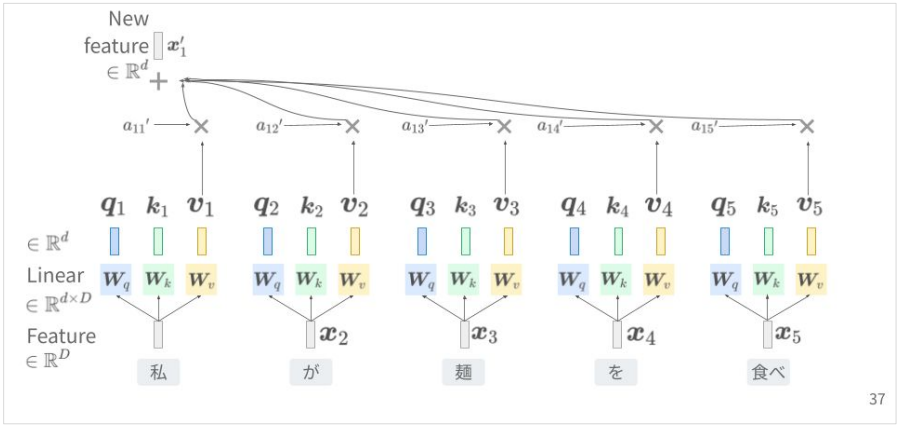
Encode Attention Decode

Self attention (slides from previous class)





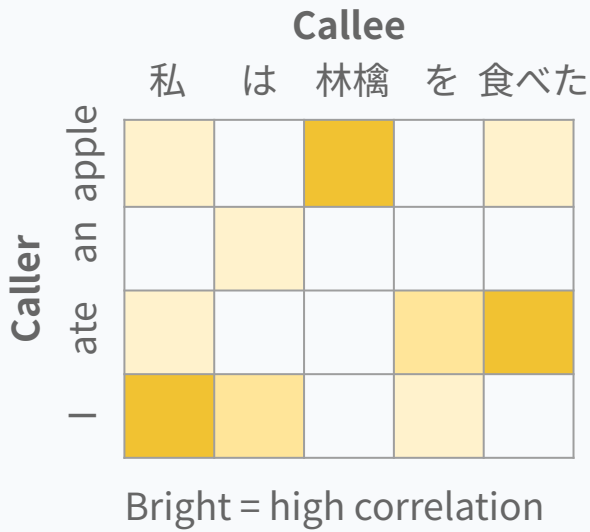




Self attention and cross attention

Cross attention

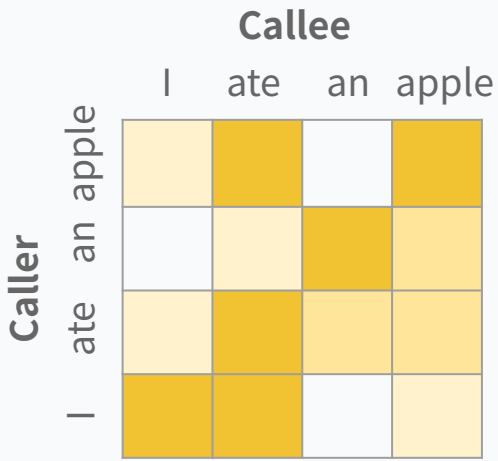
Correlation between two sequences.



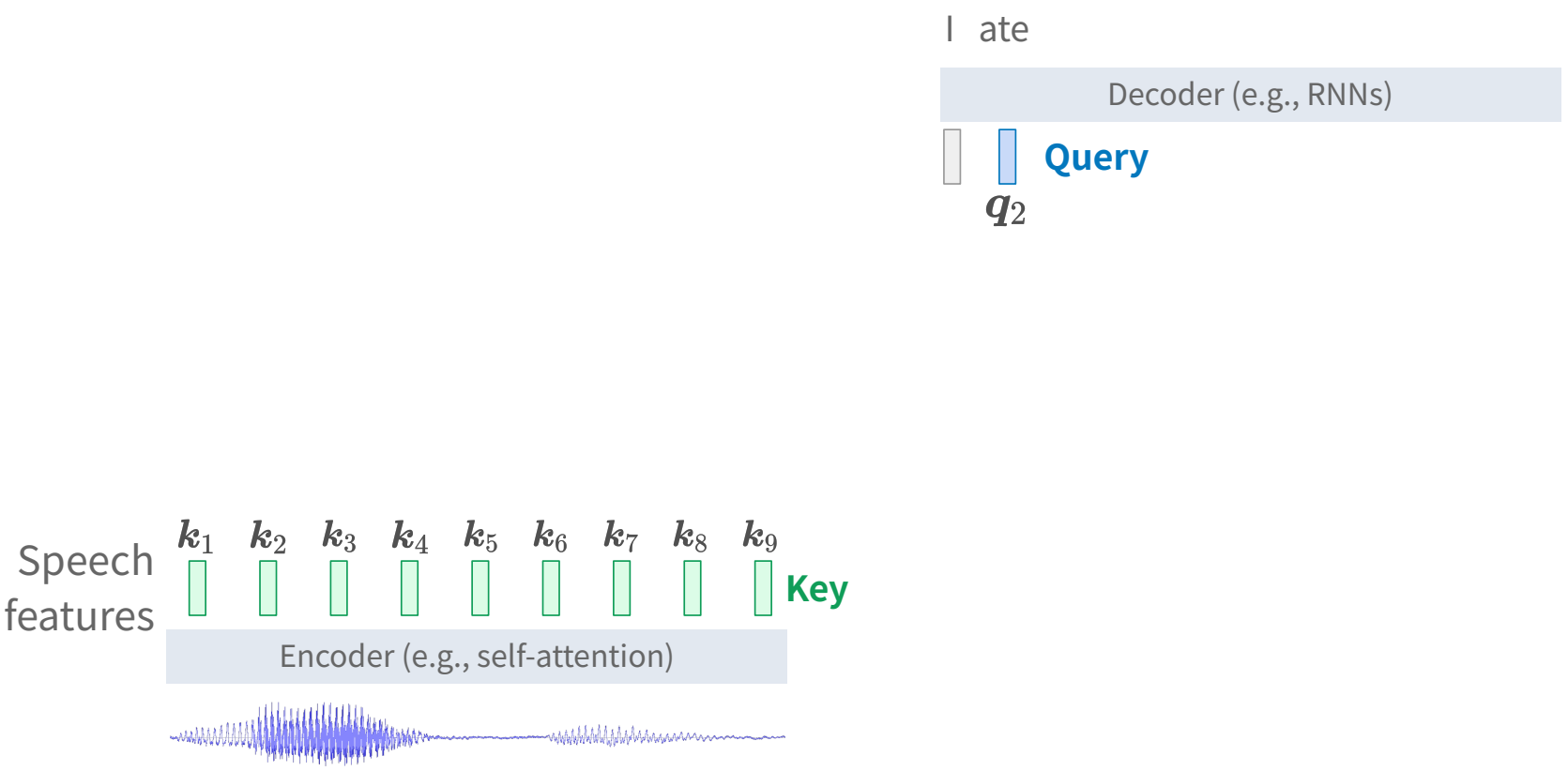
Bright = high correlation

Self attention

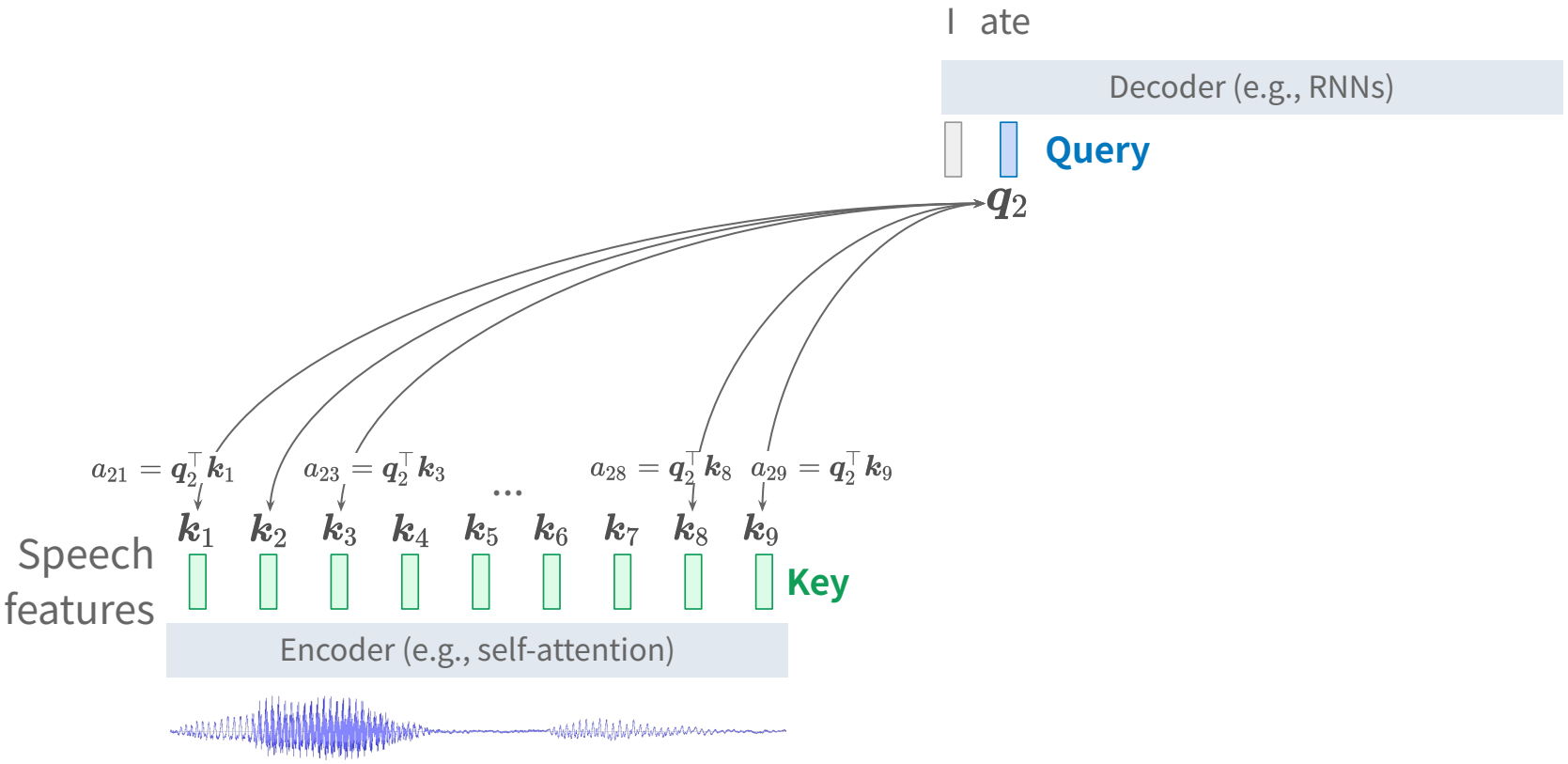
Correlation in one sequence.



Text side is caller, speech side is callee.



Inner product to calculate correspondence

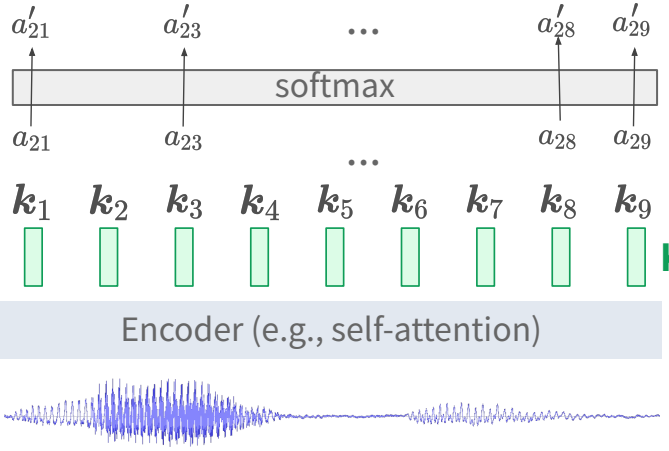


Apply softmax to be “attention weights”

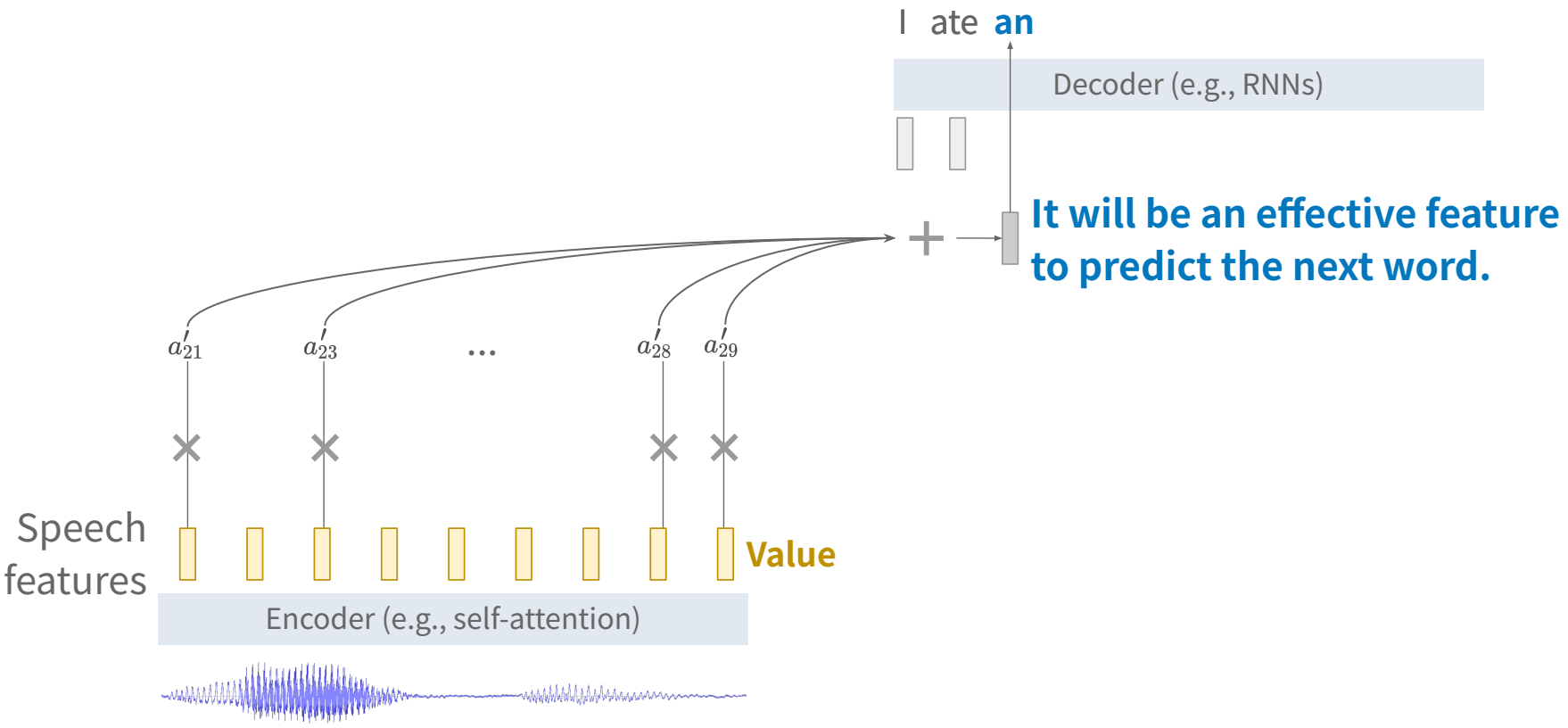
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Decoder (e.g., RNNs)

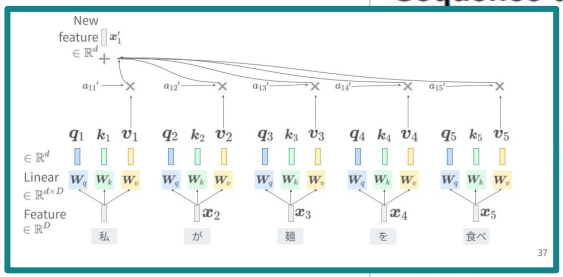
Normalized correspondance



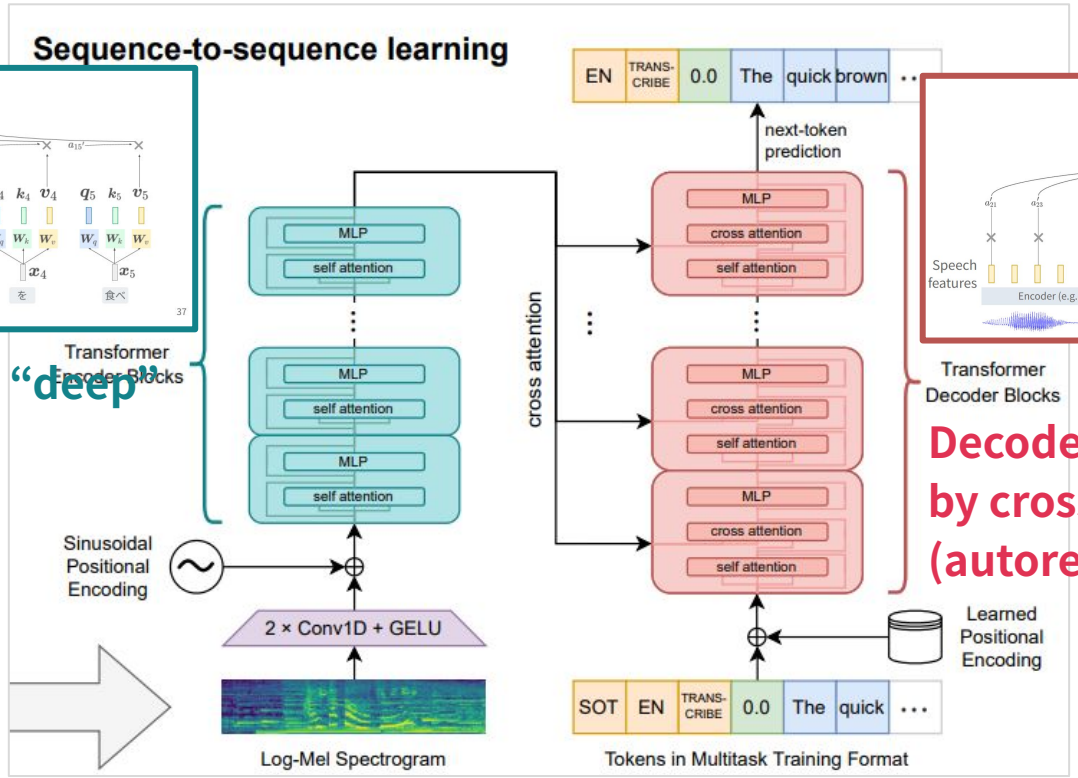
Weighted values become a new feature!

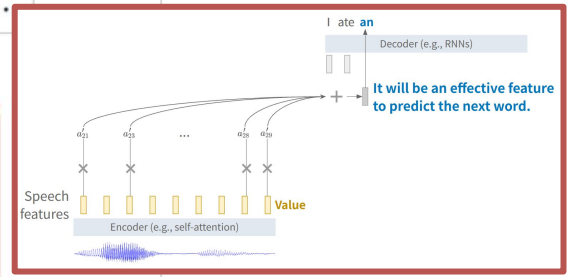


Whisper is based on LAS



Encoder extracts "deep" features by self-attention



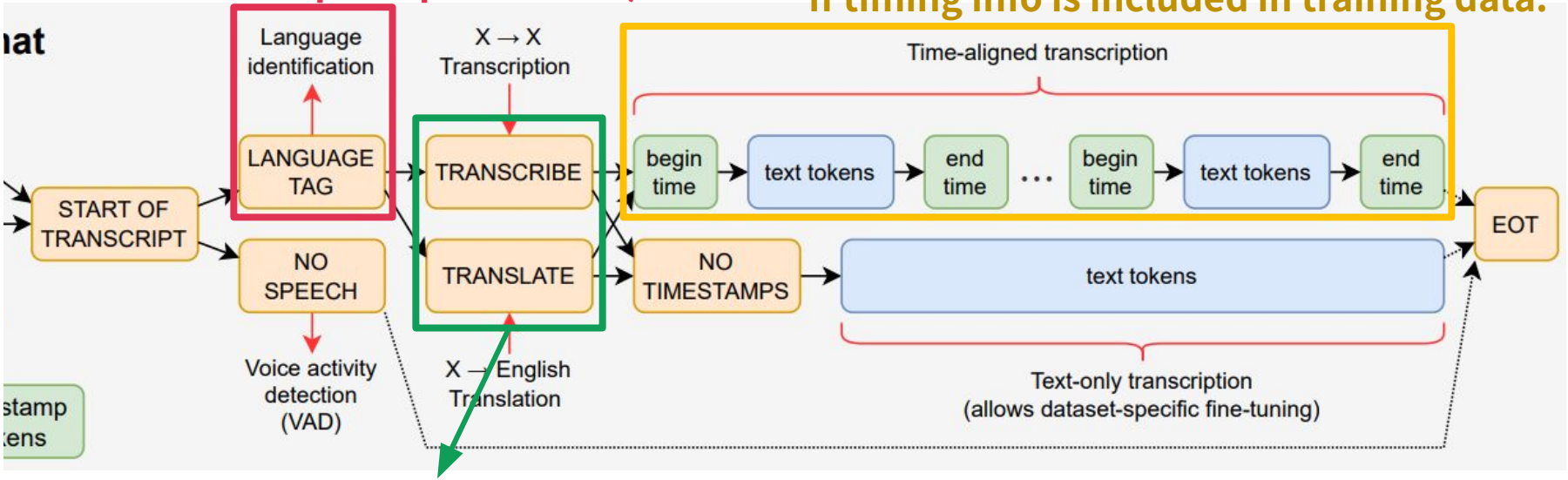


Decoder predicts words by cross-attention (autoregressive manner)

Whisper-style multi-task learning

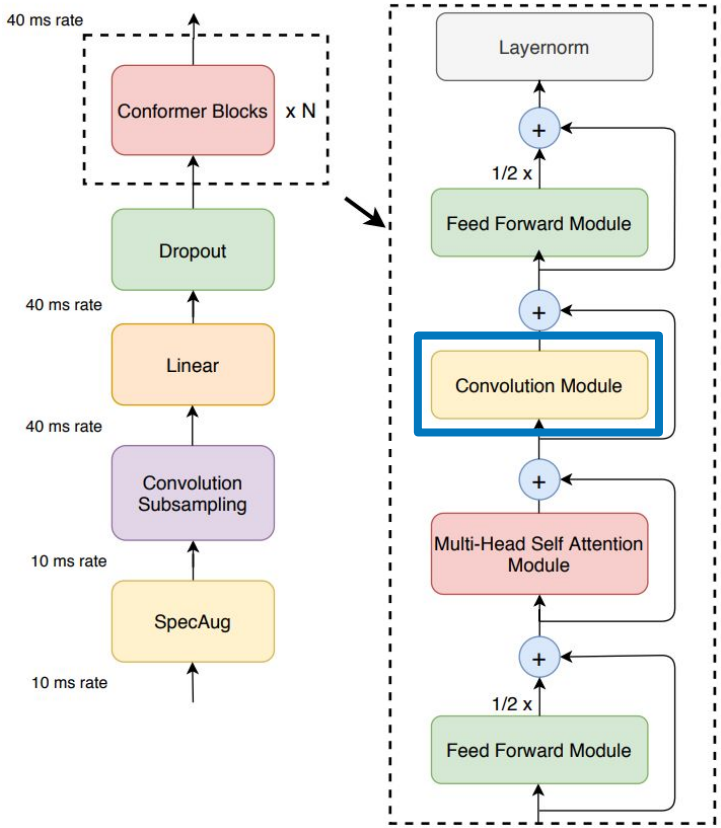
Firstly identifies language spoken, e.g., En, Ja, etc. (for guiding subsequent predictions)

Predicts timings and transcription If timing info is included in training data.



Uses speech-text data or speech-(translated text) data for training

Conformer: Convolution-augmented Transformer



Transformer + CNN -> **Conformer**

- Transformer captures long contexts, e.g., prosody
- CNN captures short contexts, phoneme

Method	#Params (M)	WER Without LM		WER With LM	
		testclean	testother	testclean	testother
Hybrid					
Transformer [33]	-	-	-	2.26	4.85
CTC					
QuartzNet [9]	19	3.90	11.28	2.69	7.25
LAS					
Transformer [34]	270	2.89	6.98	2.33	5.17
Transformer [19]	-	2.2	5.6	2.6	5.7
LSTM	360	2.6	6.0	2.2	5.2
Transducer					
Transformer [7]	139	2.4	5.6	2.0	4.6
ContextNet(S) [10]	10.8	2.9	7.0	2.3	5.5
ContextNet(M) [10]	31.4	2.4	5.4	**2.0**	4.5
ContextNet(L) [10]	112.7	**2.1**	4.6	**1.9**	4.1
Conformer (Ours)					
Conformer(S)	10.3	**2.7**	**6.3**	**2.1**	**5.0**
Conformer(M)	30.7	**2.3**	**5.0**	**2.0**	**4.3**
Conformer(L)	118.8	**2.1**	**4.3**	**1.9**	**3.9**

Evaluation

How we evaluate ASR hypothesis?

(ref: reference, Hyp: hypothesis, ASR output)

- SER (sentence error rate)
  - The sentence is totally correct or not (0% or 100%)

Ref.) 東京大学で学んでいます
Hyp.) 統計学を学んでいます

SER =100 %

- WER (word error rate)
  - Word-level edit distance (編集距離)

Ref.) 東京/大学/で/学ん/で/い/ます
Hyp.) 統計学/を/学ん/で/い/ます

Insertion error (挿入誤り)：0
Deletion error (削除誤り)：1
Substituion error (置換誤り)：2
Ref. length： 7
WER = (2 + 1) / 7 * 100 [%]

- CER (character error rate)
  - Character-level edit distance

Ref.) 東/京/大/学/で/学/ん/で/い/ま/す
Hyp.) 統/計/学/を/学/ん/で/い/ま/す

In the same manner as in WER

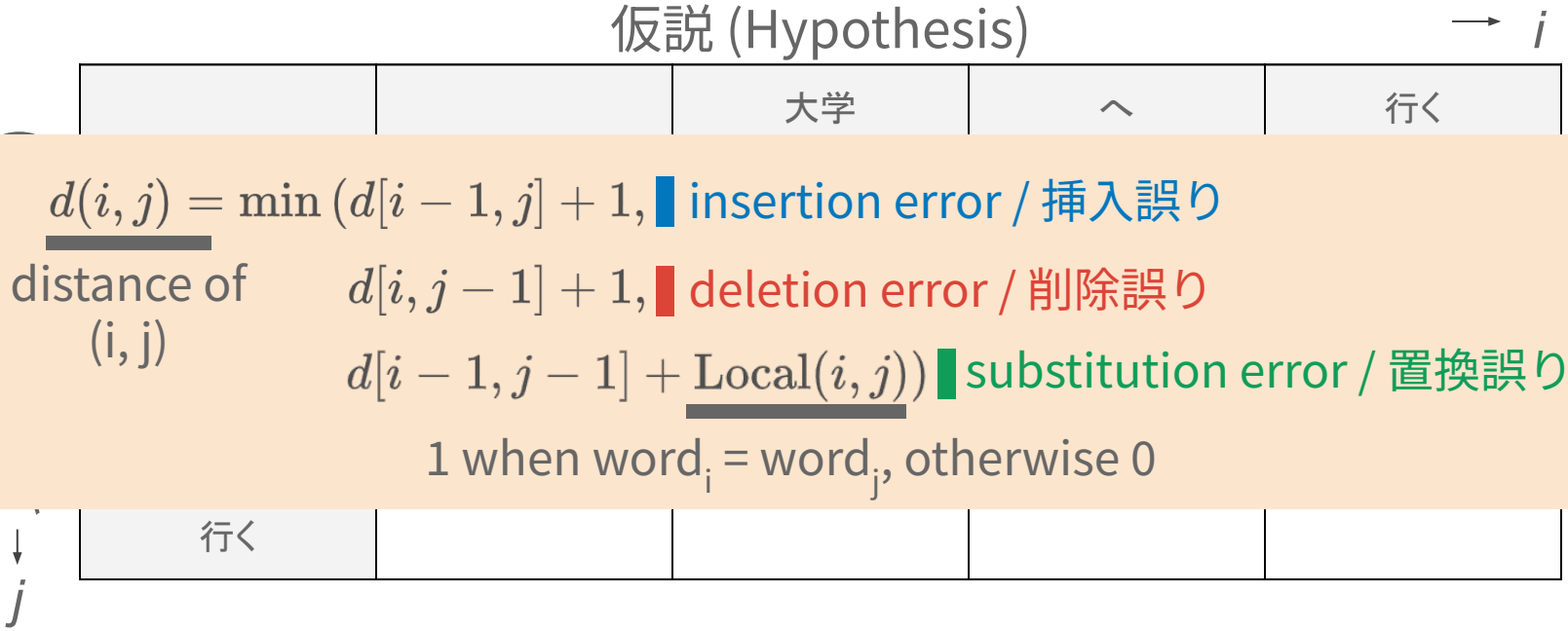
WER calculation based on Levenshtein distance /レーベンシュタイン距離に基づく WER の計算

仮説 (Hypothesis) → *i*

↓ *j* 見本 (Reference)		大学	へ	行く
きょう		?		
カーニバル				
へ				
行く				

WER is the edit distance of two sequences of diff. lengths that have been aligned.

WER calculation based on Levenshtein distance /レーベンシュタイン距離に基づく WER の計算



$$d(i,j) = \min\left(d[i-1,j]+1,\ d[i,j-1]+1,\ d[i-1,j-1]+\mathrm{Local}(i,j)\right)$$

distance of (i, j)

- $d[i-1,j]+1$: insertion error / 挿入誤り
- $d[i,j-1]+1$: deletion error / 削除誤り
- $d[i-1,j-1]+\mathrm{Local}(i,j)$: substitution error / 置換誤り

1 when $word_i = word_j$, otherwise 0

WER is the edit distance of two sequences of diff. lengths that have been aligned.

WER calculation based on Levenshtein distance /レーベンシュタイン距離に基づく WER の計算

仮説 (Hypothesis) → i

		大学	へ	行く
	0 →	**1** →	**2** →	**3**
きょう				
カーニバル				
へ				
行く				

j ↓ 見本 (Reference)

$$\min\left(d[i-1,j]+1,\ d[i,j-1]+1,\ d[i-1,j-1]+\mathrm{Local}(i,j)\right)$$

Insertion / 挿入　Deletion / 削除　Substitution / 置換

WER calculation based on Levenshtein distance /レーベンシュタイン距離に基づく WER の計算

仮説 (Hypothesis) → i

↓ j 見本 (Reference)		大学	へ	行く
	0	1	2	3
きょう	1			
カーニバル	2			
へ	3			
行く	4			

$$\min\left(d[i-1,j]+1,\ d[i,j-1]+1,\ d[i-1,j-1]+\mathrm{Local}(i,j)\right)$$

Insertion / 挿入　Deletion / 削除　Substitution / 置換

WER calculation based on Levenshtein distance /レーベンシュタイン距離に基づく WER の計算

仮説 (Hypothesis) → i

		大学	へ	行く
	0	1	2	3
きょう	1	min(2, 2, 1)		
カーニバル	2			
へ	3			
行く	4			

見本 (Reference) ↓ j

$$\min\left(d[i-1,j]+1, d[i,j-1]+1, d[i-1,j-1]+\mathrm{Local}(i,j)\right)$$

Insertion / 挿入　Deletion / 削除　Substitution / 置換

WER calculation based on Levenshtein distance /レーベンシュタイン距離に基づく WER の計算

仮説 (Hypothesis) → i

		大学	へ	行く
	0	1	2	3
きょう	1	1		
カーニバル	2	min(3, 2, 2)		
へ	3			
行く	4			

↓ 見本 (Reference) j

$$\min\left(d[i-1,j]+1, d[i,j-1]+1, d[i-1,j-1]+\mathrm{Local}(i,j)\right)$$

Insertion / 挿入　Deletion / 削除　Substitution / 置換

WER calculation based on Levenshtein distance /レーベンシュタイン距離に基づく WER の計算

仮説 (Hypothesis) → i

		大学	へ	行く
	0	1	2	3
きょう	1	1	2	
カーニバル	2	2	2	
へ	3	3	min(4, 3, 2) Local(i,j)=0	
行く	4	4		

j ↓ 見本 (Reference)

$$\min\left(d[i-1,j]+1, d[i,j-1]+1, d[i-1,j-1]+\mathrm{Local}(i,j)\right)$$

Insertion / 挿入　Deletion / 削除　Substitution / 置換

WER calculation based on Levenshtein distance /レーベンシュタイン距離に基づく WER の計算

仮説 (Hypothesis) → i

↓ j 見本 (Reference)		大学	へ	行く
	0	1	2	3
きょう	1	1	2	3
カーニバル	2	2	2	3
へ	3	3	2	3
行く	4	4	3	2

$$\min\left(d[i-1,j]+1, d[i,j-1]+1, d[i-1,j-1]+\mathrm{Local}(i,j)\right)$$

Insertion / 挿入　Deletion / 削除　Substitution / 置換

WER calculation based on Levenshtein distance /レーベンシュタイン距離に基づく WER の計算

仮説 (Hypothesis) → i

見本 (Reference) ↓ j		大学	へ	行く
	0	1	2	3
きょう	1	1	2	3
カーニバル	2	2	2	3
へ	3	3	2	3
行く	4	4	3	2

The shortest path is their alignment.
（Reference length = 4, insertion = 0, deletion = 1, substitution = 1）

WER calculation based on Levenshtein distance /レーベンシュタイン距離に基づく WER の計算

- WER (CER) may exceed 100%.
  - E.g., Many insertion and deletion.

- Alignment path is not unique.
  - Multiple shortest paths may exist. One path is selected on some criteria.

- Check not only WER but also inner errors.
  - Many insertion errors … failed to detect end of speech.
  - Many deletion errors … much background noises.

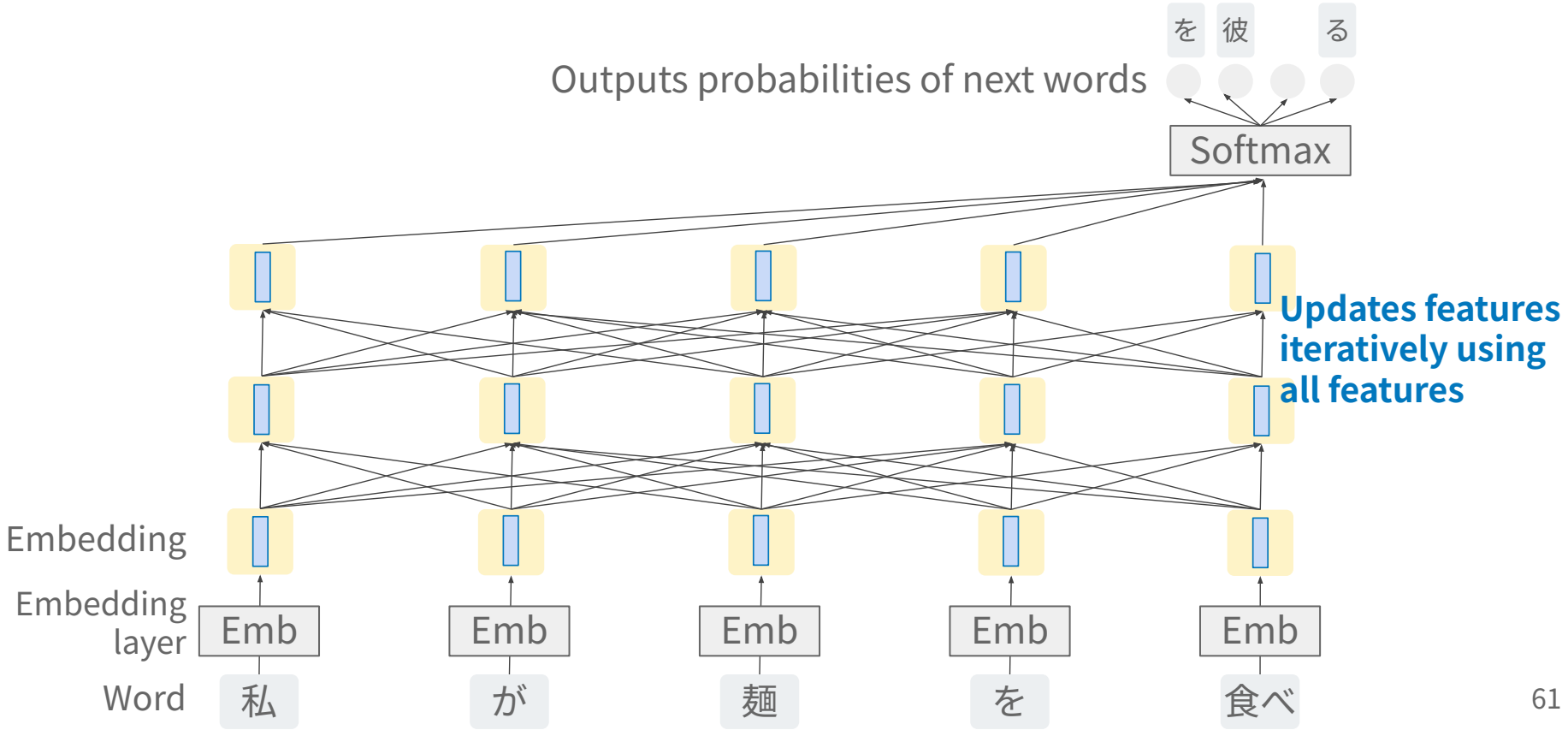
Advanced metrics

- Speed & Price
  - RTFx (real-time factor x): speech duration / processing time
  - US doller per 1000-minute speech

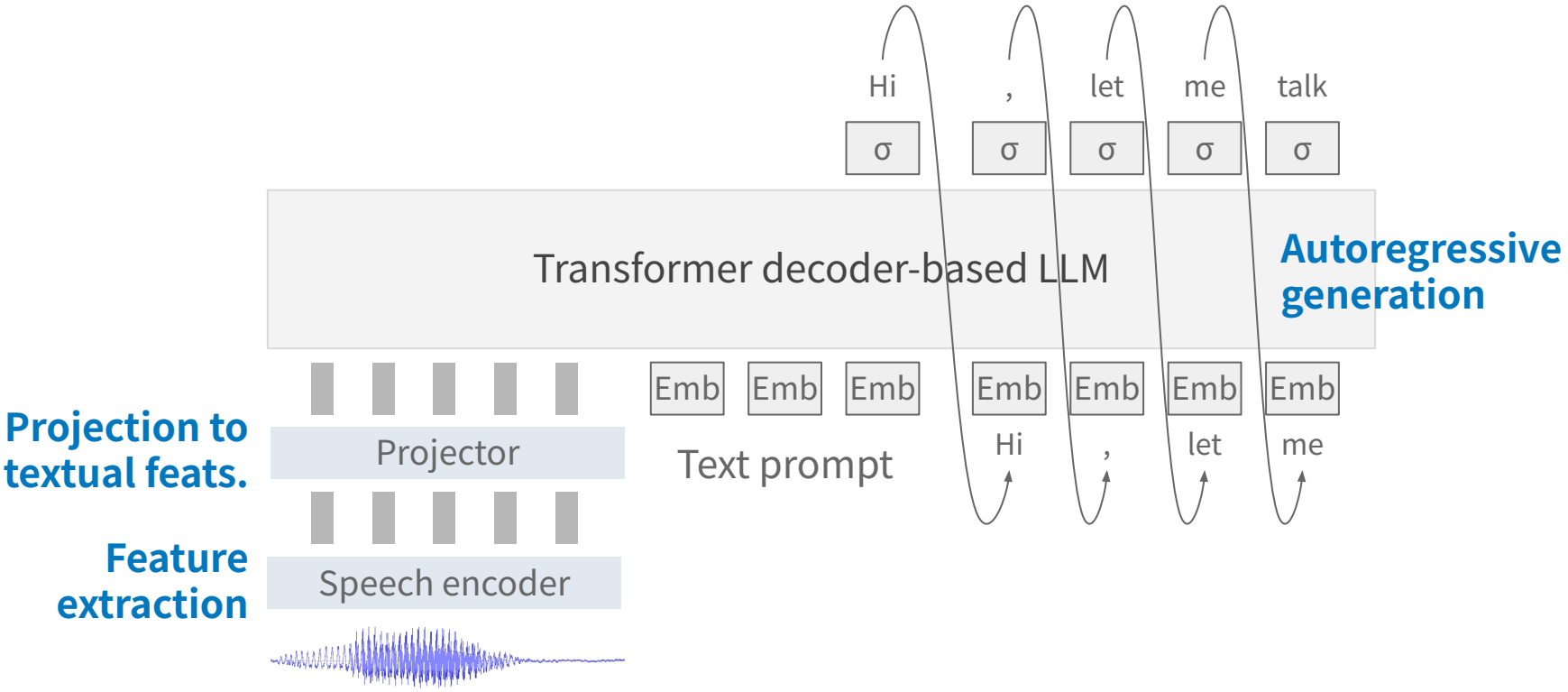
- Semantic metrics:
  - WER just considers “textual” mismatch.
  - BERTScore: distance between BERT features

LLM-based ASR / 大規模言語モデルを用いたASR

Revisit: Inference of LLMs



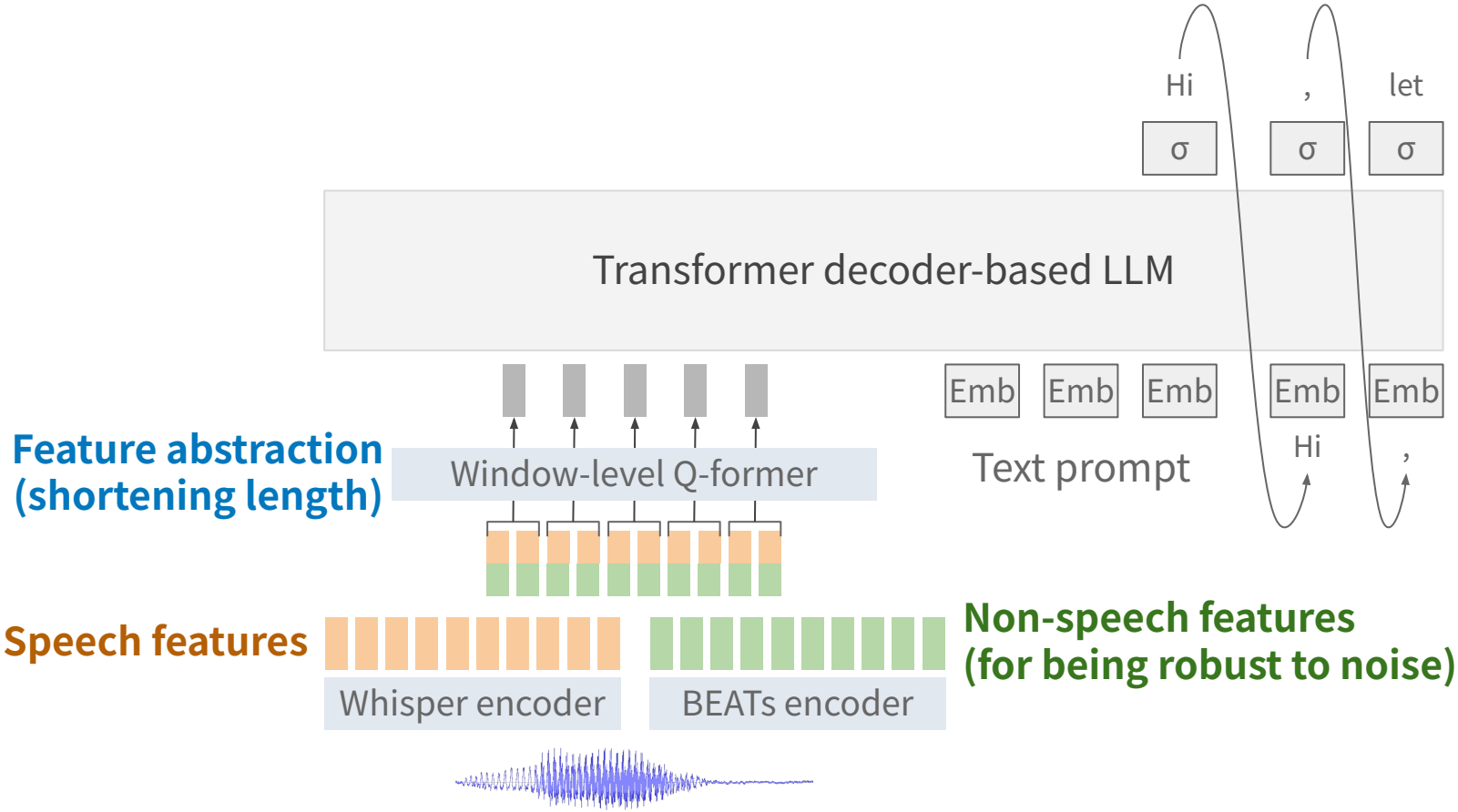
Typical architecture of LLM-based ASR



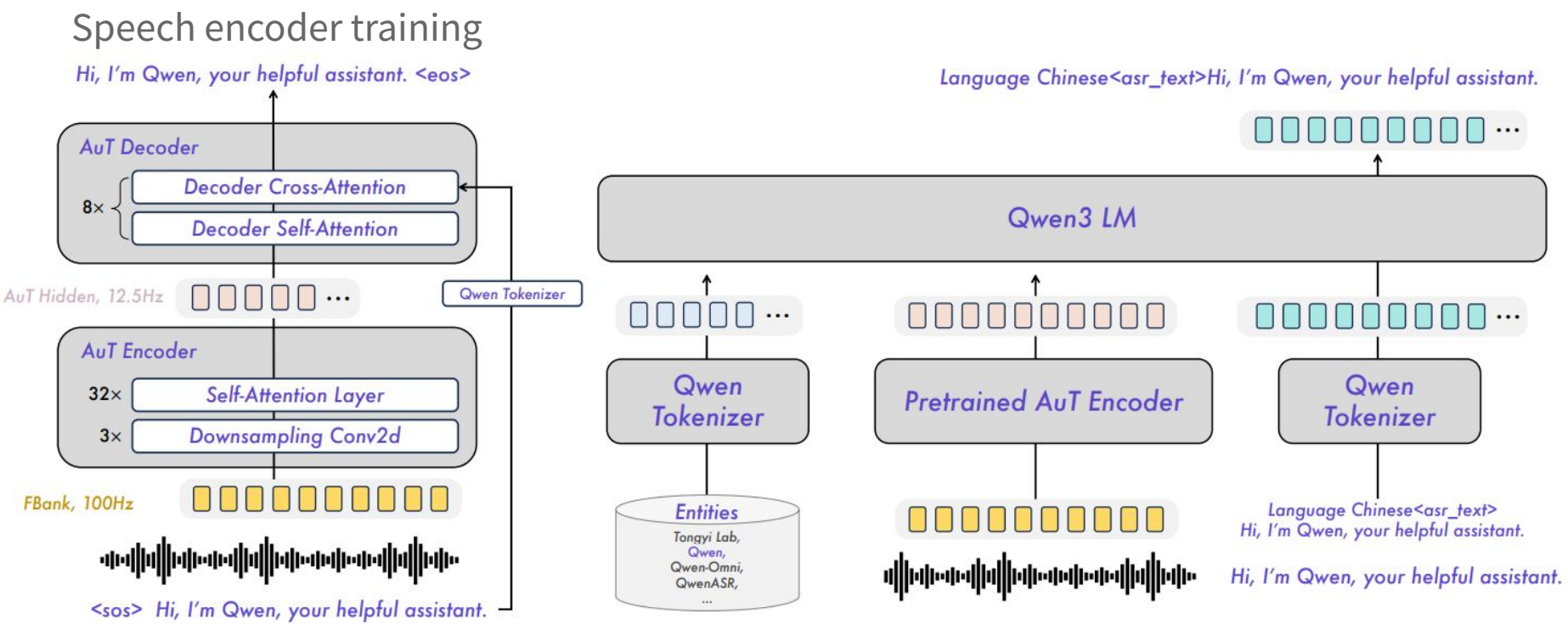
Why LLMs for ASR?

- **Language knowledge:** Next-token prediction earns abilities of generating natural text.
- **Language adaptation:** Pretrained models easily adapt to new languages, using pre-trained language data.
- **Scalability:** Data size and model size (in log scale) affect to the performance linearly.

SALMONN: dual encoder & Q-former (2023)



Qwen3-ASR (2026)



Conclusion / まとめ

Conclusion

- Conclusion
  - History, Formulation
  - CTC, LAS, and LLM-based
  - Evaluation

- Many tasks are not solved:
  - Multi-talker ASR: who speaks (when &) what?
  - Low-resource language ASR: as of now, only ASR supports only 100 languages out of 7000 languages.
  - On-device ASR: available on microcomputer?
  - Long context (60 min~) ASR: consistency in contexts?
  - Contextual ASR: domain- or conversation-specific words

Next week: guest talk



- Taishi Nakashima / 中嶋 大志, Ph.D
  - Earned his degree in 2024.
- Assistant Professor, Tokyo Metropolitan University / 東京都立大学 助教
- Acoustic signal processing / 音響信号処理
- https://taishi.org/

Slides in English, Talk in Japanese (w/ Japanese-to-English translation on Zoom)